7-7

1. Let f be the function satisfying $f'(x) = 4x - 2x\,f(x)$ for all real numbers x, with $f(0) = 5$ and $\lim_{x\to\infty} f(x) = 2$.

Find the particular solution $y = f(x)$ to the differential equation $dy/dx = 4x - 2xy$ with the initial condition $f(0) = 5$.

Part C

One point is earned for separate variables

One point is earned for antiderivatives

One point is earned for constant of integration

One point is earned for uses initial condition

One point is earned for solves for y

$$\frac{1}{4-2y}dy = xdx$$

$$-\frac{1}{2}\ln|4-2y| = \frac{1}{2}x^2 + A$$

$$\ln|4-2y| = -x^2 + B$$

$$2y - 4 = Ce^{-x^2}$$

$$C = 6$$

$$y = 2 + 3e^{-x^2}$$

for all real numbers x

0	1	2	3	4	5

The student earns all of the following points:

One point is earned for separate variables

7-7

One point is earned for antiderivatives

One point is earned for constant of integration

One point is earned for uses initial condition

One point is earned for solves for y

$$\frac{1}{4-2y}dy = xdx$$

$$-\frac{1}{2}\ln \mid 4-2y \mid = \frac{1}{2}x^2 + A$$

$$\ln \mid 4-2y \mid = -x^2 + B$$

$$2y - 4 = Ce^{-x^2}$$

$$C = 6$$

$$y = 2 + 3e^{-x^2}$$

for all real numbers x

7-7

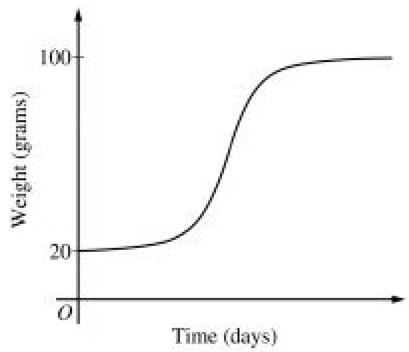
2. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

(a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

(b) Find $\frac{d^2B}{dt^2}$ in terms of B. Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



(c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left.\frac{dB}{dt}\right|_{B=40} = \frac{1}{5}(60) = 12$$
$$\left.\frac{dB}{dt}\right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left.\frac{dB}{dt}\right|_{B=40} > \left.\frac{dB}{dt}\right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

0	1	2

The student earns all of the following points:

7-7

1 point is earned for using $\dfrac{dB}{dt}$

1 point is earned for the answer with reason

$$\left.\frac{dB}{dt}\right|_{B=40} = \tfrac{1}{5}(60) = 12$$
$$\left.\frac{dB}{dt}\right|_{B=70} = \tfrac{1}{5}(30) = 6$$

Because $\left.\dfrac{dB}{dt}\right|_{B=40} > \left.\dfrac{dB}{dt}\right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

Part B

1 point is earned for the $\dfrac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100-B) = -\frac{1}{25}(100-B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

0	1	2 ✓

The student earns all of the following points:

1 point is earned for the $\dfrac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100-B) = -\frac{1}{25}(100-B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

Part C

7-7

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$
$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$
$$-\ln|100 - B| = \frac{1}{5}t + c$$
Because $20 \leq B < 100, \ |100 - B| = 100 - B.$
$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$
$$100 - B = 80e^{-t/5}$$
$$B(t) = 100 - 80e^{-t/5}, \ t \geq 0$$

0	1	2	3	4	5

The student earns all of the following points:

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

7-7

$$\frac{dB}{dt} = \tfrac{1}{5}(100 - B)$$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \tfrac{1}{5}t + c$$

Because $20 \le B < 100,\ |100 - B| = 100 - B.$

$$-\ln(100 - 20) = \tfrac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5},\ t \ge 0$$

7-7

3. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then $\frac{db}{dt} = \frac{1}{5}(100 - B)$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part C

One point is earned for the separation of variables

One point is earned for the anitderivatives

One point is earned for the constant of integration

One point is earned for uses initial condition

One point is earned for solves for B

$$\frac{dB}{dt}| = \frac{1}{5}(100 - B)$$
$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$
$$-\ln|100 - B| = \frac{1}{5}t + c$$
$$Because, 20 \le B \le 100, |100 - B| = 100 - B$$
$$-\ln|100 - 20| = \frac{1}{5}(0) + c \Rightarrow -\ln(80) = C$$
$$100 - B = 80e^{-t/5}$$
$$B(t) = 100 - 80e^{-t/5}, t \ge 0$$

0	1	2	3	4	5

The student earns all of the following points:

One point is earned for the separation of variables

One point is earned for the anitderivatives

One point is earned for the constant of integration

One point is earned for uses initial condition

One point is earned for solves for B

7-7

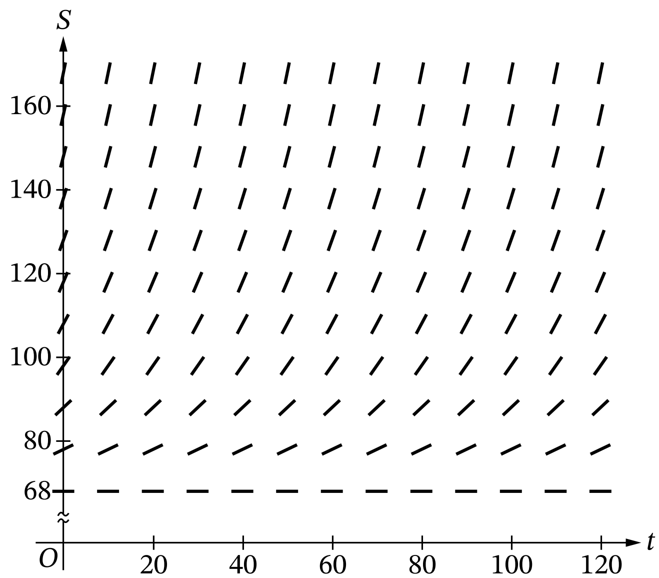
$$\frac{dB}{dt}\Big|= \tfrac{1}{5}(100 - B)$$
$$\int \frac{1}{100 - B}dB = \int \frac{1}{5}dt$$
$$- \ln|100 - B| = \tfrac{1}{5}t + c$$
$$Because, 20 \le B \le 100, |100 - B| = 100 - B$$
$$- \ln|100 - 20| = \tfrac{1}{5}(0) + c \Rightarrow - \ln(80) = C$$
$$100 - B = 80e^{-t/5}$$
$$B(t) = 100 - 80e^{-t/5}, t \ge 0$$

7-7

4. For $0 \leq t \leq 120$, the temperature of a bowl of soup at time t is modeled by the function S that satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$, where $S(t)$ is measured in degrees Fahrenheit ($^\circ$F) and t is measured in minutes. The temperature of the bowl of soup is always greater than 68°F and is 148°F at time $t = 0$.

Part A

Explain why the following could not be a slope field for the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$.



Part B

Use the line tangent to the graph of S at $t = 0$ to approximate the temperature of the soup at time $t = 5$ minutes.

Part C

According to the model, is the approximation found in part B an overestimate or an underestimate of the temperature of the soup at time $t = 5$ minutes? Give a reason for your answer.

Part D

Using separation of variables, find the particular solution $y = S(t)$ to the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$ with initial condition $S(0) = 148$.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

7-7

- **P1** can be earned for a response of "because there are no negative slopes" or "because the slopes are positive."
- **P1** can be earned by using a local argument at any point (t, S) with $0 \leq t \leq 120$ and $S > 68$, rather than a global argument.

Model Solution:

The slopes of the line segments in a slope field for the given differential equation should all be negative because for $S > 68$, $\frac{dS}{dt} < 0$. However, the slopes of the line segments in the slope field given are all nonnegative.

0	1

The response accurately includes the criteria below.

- Explanation

Part B Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response can earn **P2** with $\frac{dS}{dt}\Big|_{(0,\ 148)} = -4$, "slope is -4," or equivalent.
- A response that presents a linear approximation with a slope of -4 also earns **P2**. For example, a response of $148 - 4(5)$ earns **P2**.
- A response that declares the slope equal to any nonzero value $k \neq -4$ does not earn **P2**.

Model Solution:

$$\frac{dS}{dt}\Big|_{(0,148)} = -\frac{1}{20}(148 - 68) = -4$$

0	1

The response accurately includes the criteria below.

- Slope of tangent line

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

7-7

Scoring Notes:

- A response that declares the slope equal to any nonzero value $k \neq -4$:
  - Earns **P3** if there is evidence that the given differential equation was used to calculate k and that k was used correctly in a tangent line approximation of the form $148 + k(5)$. Subsequent errors in simplification will not be considered in scoring for **P3**.
  - Does not earn **P3** if k is simply declared to be a slope without a connection to the differential equation.
- **P3** cannot be earned for an approximation at any value of t other than 5.
- A response does not have to present the tangent line equation but must clearly demonstrate its use at $t = 5$ in finding the requested approximation in order to earn **P3**.
- A response of $148 - 4(5)$ earns **P3**. Subsequent errors in simplification will not be considered in scoring for **P3**.

Model Solution:

$S(5) \approx 148 - 4(5) = 128$

The temperature of the soup at $t = 5$ minutes is approximately $128^\circ\mathrm{F}$.

0	1

The response accurately includes the criteria below.

- Tangent line approximation

Part C Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, a response must include one of the following: $\frac{d^2S}{dt^2}$, $-\frac{1}{20}\frac{dS}{dt}$, $\frac{1}{400}(S - 68)$, “the second derivative,” or equivalent.
  - “S' is increasing (or decreasing)” or a reference to concavity without a connection to the second derivative is not sufficient to earn **P4**.

Model Solution:

Because $S(t) > 68$, $\frac{d^2S}{dt^2} = -\frac{1}{20}\frac{dS}{dt} = \left(-\frac{1}{20}\right)\left(-\frac{1}{20}\right)(S - 68) = \frac{1}{400}(S - 68) > 0$.

7-7

0	1

The response accurately includes the criteria below.

- Considers $\frac{d^2S}{dt^2}$

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible to earn **P5**, a response must have earned **P4**.
- Responses do not need to say "concave up" to earn **P5**. A response of "underestimate because the second derivative is positive" is sufficient to earn **P5**.

Model Solution:

Because the second derivative is always positive, the tangent line approximation is an underestimate of the temperature predicted by the model.

0	1

The response accurately includes the criteria below.

- Answer with reason

Part D Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P6**.
- A response earns **P6** if the separation of variables results in an equation of the form $A\frac{dS}{S-68} = B\ dt$, where A and B are nonzero constants.
- Special case: If the response begins with a separation of the form $A\frac{dS}{S+68} = B\ dt$, where A and B are correct coefficients, it earns **P6**.

Model Solution:

7-7

$$\frac{dS}{S-68} = -\frac{1}{20}dt$$

0	1

The response accurately includes the criteria below.

- Separate variables

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P7**.
- A response is eligible to earn **P7** if the separation of variables resulted in an equation of the form $A\frac{dS}{S-68} = B\ dt$, where A and B are nonzero constants. A response that presents consistent antiderivatives of the form $A\ln(S-68)$ (with either parentheses or absolute value) and Bt earns **P7**.
- Special case: If the response began with a separation of the form $A\frac{dS}{S+68} = B\ dt$, where A and B are correct coefficients, it earns **P7** with two consistent antiderivatives.
- A response with no constant of integration is eligible to earn **P7**.

Model Solution:

$$\ln|S-68| = -\frac{t}{20} + C$$

0	1

The response accurately includes the criteria below.

- Finds antiderivatives

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P8**.
- A response with no constant of integration does not earn **P8**.

7-7

- A response is eligible for **P8** if it has earned P6 and has one of the two antiderivatives correct OR if it has earned **P7**.
- An eligible response earns **P8** by correctly including the constant of integration in an equation and substituting 0 for t and 148 for S.

Model Solution:

$$\ln|148-68| = -\frac{0}{20} + C \Rightarrow \ln 80 = C$$

$$\ln(S-68) = -\frac{t}{20} + \ln 80$$

0	1

The response accurately includes the criteria below.

- Constant of integration and uses initial condition

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response is eligible for **P9** only if it has earned **P6**, **P7**, and **P8**.
- A response earns **P9** only for an answer of $S(t) = 68 + 80e^{-t/20}$ or equivalent.

Model Solution:

$$S - 68 = e^{-t/20+\ln 80} = 80e^{-t/20}$$

$$S(t) = 68 + 80e^{-t/20}$$

0	1

The response accurately includes the criteria below.

- Solves for $S(t)$

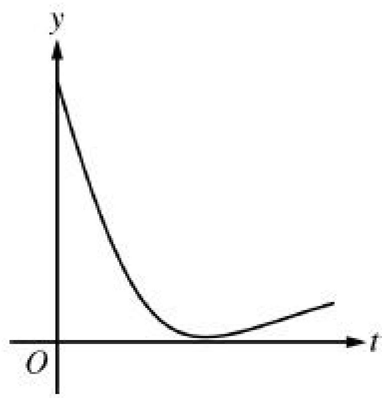
7-7

5. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

During a chemical reaction, the function $y = f(t)$ models the amount of a substance present, in grams, at time t seconds. At the start of the reaction $(t = 0)$, there are 10 grams of the substance present. The function $y = f(t)$ satisfies the differential equation $\frac{dy}{dt} = -0.02y^2$.

(a) Use the line tangent to the graph of $y = f(t)$ at $t = 0$ to approximate the amount of the substance remaining at time $t = 2$ seconds.

(b) Using the given differential equation, determine whether the graph of f could resemble the following graph. Give a reason for your answer.



(c) Find an expression for $y = f(t)$ by solving the differential equation $\frac{dy}{dt} = -0.02y^2$ with the initial condition $f(0) = 10$.

(d) Determine whether the amount of the substance is changing at an increasing or a decreasing rate. Explain your reasoning.

Part A

1 point is earned for: $y'(0)$

1 point is earned for: approximation

$y'(0) = -0.02\left(10^2\right) = -2$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$y(2) \approx 10 - 2(2) = 6$ grams

0	1	2

The student response earns all of the following points:

1 point is earned for: $y'(0)$

1 point is earned for: approximation

7-7

$y'(0) = -0.02\left(10^2\right) = -2$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$y(2) \approx 10 - 2(2) = 6$ grams

Part B

1 point is earned for: answer with reason

$\frac{dy}{dt} = -0.02y^2 \leq 0$, so the graph of *f* is nonincreasing.

The graph of *f* cannot resemble the given graph because the given graph is increasing on a portion of its domain.

0	1

The student response earns all of the following points:

1 point is earned for: answer with reason

$\frac{dy}{dt} = -0.02y^2 \leq 0$, so the graph of *f* is nonincreasing.

The graph of *f* cannot resemble the given graph because the given graph is increasing on a portion of its domain.

Part C

1 point is earned for: separation of variables

1 point is earned for: antiderivatives

1 point is earned for: constant of integration and uses initial condition

1 point is earned for: answer

$$\int \left(-\frac{1}{y^2}\right) dy = \int 0.02\, dt$$

$$\frac{1}{y} = 0.02t + C$$

$$\frac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$$

$$\frac{1}{y} = 0.02t + 0.1 \Rightarrow y = \frac{1}{0.02t+0.1} = \frac{50}{t+5}$$

Note: this solution is valid for $t > -5$.

7-7

Note: max $2/4$ $[1\text{-}1\text{-}0\text{-}0]$ if no constant of integration

Note: $0/4$ if no separation of variables

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for: separation of variables

1 point is earned for: antiderivatives

1 point is earned for: constant of integration and uses initial condition

1 point is earned for: answer

$$\int \left(-\frac{1}{y^2}\right) dy = \int 0.02\, dt$$
$$\frac{1}{y} = 0.02t + C$$
$$\frac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$$
$$\frac{1}{y} = 0.02t + 0.1 \Rightarrow y = \frac{1}{0.02t+0.1} = \frac{50}{t+5}$$

Note: this solution is valid for $t > -5$.

Note: max $2/4$ $[1\text{-}1\text{-}0\text{-}0]$ if no constant of integration

Note: $0/4$ if no separation of variables

Part D

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned}\frac{d^2y}{dt^2} &= -0.04y\frac{dy}{dt}\\ &= -\,0.04y(-0.02y^2)\\ &= 0.0008y^3\end{aligned}$$

Because $y > 0$, $0.0008y^3 > 0$.

The amount of substance is changing at an increasing rate.

— OR —

7-7

From part (c), $f(t) = \frac{50}{t+5}$, and from context, $t \geq 0$.

$f'(t) = \frac{-50}{(t+5)^2}$ and $f''(t) = \frac{100}{(t+5)^3} > 0$ for $t \geq 0$.

The amount of substance is changing at an increasing rate.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned} \frac{d^2y}{dt^2} &= -0.04y\frac{dy}{dt} \\ &= -0.04y(-0.02y^2) \\ &= 0.0008y^3 \end{aligned}$$

Because $y > 0$, $0.0008y^3 > 0$.

The amount of substance is changing at an increasing rate.

— OR —

From part (c), $f(t) = \frac{50}{t+5}$, and from context, $t \geq 0$.

$f'(t) = \frac{-50}{(t+5)^2}$ and $f''(t) = \frac{100}{(t+5)^3} > 0$ for $t \geq 0$.

The amount of substance is changing at an increasing rate.

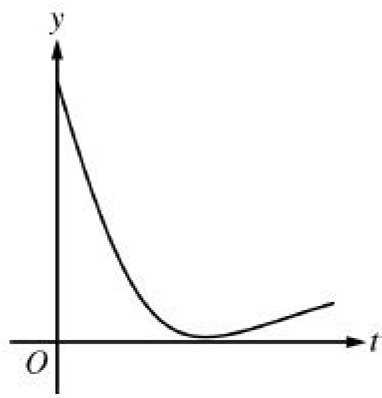
7-7

6. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

During a chemical reaction, the function $y = f(t)$ models the amount of a substance present, in grams, at time t seconds. At the start of the reaction $(t = 0)$, there are 10 grams of the substance present. The function $y = f(t)$ satisfies the differential equation $\frac{dy}{dt} = -0.02y^2$.

(a) Use the line tangent to the graph of $y = f(t)$ at $t = 0$ to approximate the amount of the substance remaining at time $t = 2$ seconds.

(b) Using the given differential equation, determine whether the graph of f could resemble the following graph. Give a reason for your answer.



(c) Find an expression for $y = f(t)$ by solving the differential equation $\frac{dy}{dt} = -0.02y^2$ with the initial condition $f(0) = 10$.

(d) Determine whether the amount of the substance is changing at an increasing or a decreasing rate. Explain your reasoning.

Part A

1 point is earned for: $y'(0)$

1 point is earned for: approximation

$y'(0) = -0.02\left(10^2\right) = -2$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$y(2) \approx 10 - 2(2) = 6$ grams

0	1	2

The student response earns all of the following points:

1 point is earned for: $y'(0)$

1 point is earned for: approximation

7-7

$y'(0) = -0.02\left(10^2\right) = -2$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$y(2) \approx 10 - 2(2) = 6$ grams

Part B

1 point is earned for: answer with reason

$\frac{dy}{dt} = -0.02y^2 \leq 0$, so the graph of *f* is nonincreasing.

The graph of *f* cannot resemble the given graph because the given graph is increasing on a portion of its domain.

0	1

The student response earns all of the following points:

1 point is earned for: answer with reason

$\frac{dy}{dt} = -0.02y^2 \leq 0$, so the graph of *f* is nonincreasing.

The graph of *f* cannot resemble the given graph because the given graph is increasing on a portion of its domain.

Part C

1 point is earned for: separation of variables

1 point is earned for: antiderivatives

1 point is earned for: constant of integration and uses initial condition

1 point is earned for: answer

$$\int \left(-\frac{1}{y^2}\right) dy = \int 0.02\ dt$$

$$\frac{1}{y} = 0.02t + C$$

$$\frac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$$

$$\frac{1}{y} = 0.02t + 0.1 \Rightarrow y = \frac{1}{0.02t+0.1} = \frac{50}{t+5}$$

Note: this solution is valid for $t > -5$.

7-7

Note: max $2/4$ [1-1-0-0] if no constant of integration

Note: $0/4$ if no separation of variables

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for: separation of variables

1 point is earned for: antiderivatives

1 point is earned for: constant of integration and uses initial condition

1 point is earned for: answer

$$\int\left(-\frac{1}{y^2}\right)dy=\int 0.02\,dt$$
$$\frac{1}{y}=0.02t+C$$
$$\frac{1}{10}=0.02(0)+C\Rightarrow C=0.1$$
$$\frac{1}{y}=0.02t+0.1\Rightarrow y=\frac{1}{0.02t+0.1}=\frac{50}{t+5}$$

Note: this solution is valid for $t>-5$.

Note: max $2/4$ [1-1-0-0] if no constant of integration

Note: $0/4$ if no separation of variables

Part D

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned}\frac{d^2y}{dt^2}&=-0.04y\frac{dy}{dt}\\&=-0.04y(-0.02y^2)\\&=0.0008y^3\end{aligned}$$

Because $y>0$, $0.0008y^3>0$.

The amount of substance is changing at an increasing rate.

— OR —

7-7

From part (c), $f(t) = \frac{50}{t+5}$, and from context, $t \geq 0$.

$f'(t) = \frac{-50}{(t+5)^2}$ and $f''(t) = \frac{100}{(t+5)^3} > 0$ for $t \geq 0$.

The amount of substance is changing at an increasing rate.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned}\frac{d^2y}{dt^2} &= -0.04y\frac{dy}{dt}\\ &= -\ 0.04y(-0.02y^2)\\ &= 0.0008y^3\end{aligned}$$

Because $y > 0$, $0.0008y^3 > 0$.

The amount of substance is changing at an increasing rate.

— OR —

From part (c), $f(t) = \frac{50}{t+5}$, and from context, $t \geq 0$.

$f'(t) = \frac{-50}{(t+5)^2}$ and $f''(t) = \frac{100}{(t+5)^3} > 0$ for $t \geq 0$.

The amount of substance is changing at an increasing rate.

7. A particle moves along the x-axis so that at any time $t > 0$, its velocity is given by $v(t) = 4 - 6t^2$. If the particle is at position $x = 7$ at $t = 1$ time, what is the position of the particle at time $t = 2$?

(A) -10

(B) -5

(C) -3

(D) 3

(E) 17

7-7

8. Let f be a function with $f(2) = -8$ such that for all points (x, y) on the graph of f, the slope given by $\frac{3x^2}{y}$.

(a) Write an equation of the line tangent to the graph of f at the point where $x = 2$ and use it to approximate $f(1.8)$.

(b) Find an expression for $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = \frac{3x^2}{y}$ with initial condition $f(2) = -8$.

Part A

The response can earn up to 3 points:

1 point earned for correct slope

$$\text{Slope} = \frac{(3)(4)}{-8} = -\frac{3}{2}$$

1 point earned for the tangent line equation

An equation for the tangent line is $y = -\frac{3}{2}(x - 2) - 8$.

1 point earned for the correct approximation

$f(1.8) \approx -\frac{3}{2}(1.8 - 2) - 8 = -7.7$

0	1	2	3 ✓

The response earns all of the following points:

1 point earned for correct slope

$$\text{Slope} = \frac{(3)(4)}{-8} = -\frac{3}{2}$$

1 point earned for the tangent line equation

An equation for the tangent line is $y = -\frac{3}{2}(x - 2) - 8$.

1 point earned for the correct approximation

$f(1.8) \approx -\frac{3}{2}(1.8 - 2) - 8 = -7.7$

7-7

Part B

The response can earn up to 6 points:

1 point earned for separation of variables

2 point earned for the antiderivatives

1 point earned for the constant of integration

1 point earned for use of the initial condition

1 point earned for the solution of y

Note: max [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

$$\int y\,dy = \int 3x^2 dx$$
$$\frac{1}{2}y^2 = x^3 + C$$
$$\frac{1}{2}(-8)^2 = 2^3 + C \Rightarrow C = 24$$
$$y^2 = 2(x^3 + 24) = 2x^3 + 48$$
$$y = -\sqrt{2x^3 + 48}$$

Note: This solution is valid for $x > -\sqrt[3]{24}$.

0	1	2	3	4	5	6

The response earns all of the following points:

1 point earned for separation of variables

2 point earned for the antiderivatives

1 point earned for the constant of integration

1 point earned for use of the initial condition

1 point earned for the solution of y

Note: max [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

7-7

$\int y\,dy = \int 3x^2 dx$

$\frac{1}{2}y^2 = x^3 + C$

$\frac{1}{2}(-8)^2 = 2^3 + C \Rightarrow C = 24$

$y^2 = 2(x^3 + 24) = 2x^3 + 48$

$y = -\sqrt{2x^3 + 48}$

Note: This solution is valid for $x > -\sqrt[3]{24}$.

7-7

9. The following are related to this scenario:

Let f be a function with $f(2) = -8$ such that for all points (x,y) on the graph of f, the slope given by $\frac{3x^2}{y}$.

Find an expression for $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = \frac{3x^2}{y}$ with initial condition $f(2) =$ -8.

Part B

The response can earn up to 6 points:

Note: 0/6 if no separation of variables

Note: max 3/6 if no constant of integration.

1 point: For separation of variables

2 point: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

$$\int ydy = \int 3x^2dx\frac{1}{2}y^2 = x^3 + C\frac{1}{2}(-8)^2 = 2^3 + C \Rightarrow C = 24y^2 = 2(x^3 + 24) = 2x^3 + 48y = -\sqrt{2x^3 + 48}$$

Note: This solution is valid for $x > -\sqrt[3]{24}$

0	1	2	3	4	5	6

The response earns all six of the following points:

1 point: For separation of variables

2 point: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

$$\int ydy = \int 3x^2dx\frac{1}{2}y^2 = x^3 + C\frac{1}{2}(-8)^2 = 2^3 + C \Rightarrow C = 24y^2 = 2(x^3 + 24) = 2x^3 + 48y = -\sqrt{2x^3 + 48}$$

Note: This solution is valid for $x > -\sqrt[3]{24}$

7-7

10. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2y$ with initial condition $f(0) = 6$. Which of the following is an expression for $f(x)$?

(A) e^{x^3}

(B) $e^{x^3} + 5$

(C) $e^{x^3} + 6$

(D) $6e^{x^3}$

Answer D

Correct. The differential equation can be solved using separation of variables and the initial condition to determine the appropriate value for the arbitrary constant.

$$\frac{dy}{dx} = 3x^2y \Rightarrow \frac{1}{y}dy = 3x^2dx$$

$$\int \frac{1}{y}dy = \int 3x^2dx \Rightarrow \ln y = x^3 + C$$

$$y = e^{x^3+C} = e^Ce^{x^3} = Ke^{x^3}$$

$$6 = Ke^0 = K$$

$$y = 6e^{x^3}$$

Note: Absolute value was not used in the antiderivative of $\frac{1}{y}$ since the initial value of y is positive.

7-7

11. Which of the following satisfies the differential equation $\frac{dy}{dx} = \cos\left(\frac{\pi}{2}x^2\right)$ with the initial condition $y(0) = 2$?

(A) $y = 2 + \sin\left(\frac{\pi}{2}x^2\right)$

(B) $y = 2 + \frac{1}{\pi x}\sin\left(\frac{\pi}{2}x^2\right)$

(C) $y = \int_2^x \cos\left(\frac{\pi}{2}t^2\right) dt$

(D) $y = 2 + \int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt$ ✓

Answer D

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivative satisfy the differential equation and initial condition. If $y = 2 + \int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt$, then $y(0) = 2 + \int_0^0 \cos\left(\frac{\pi}{2}t^2\right) dt = 2 + 0 = 2$ and $\frac{dy}{dx} = \frac{d}{dx}\left(2 + \int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt\right) = \frac{d}{dx}\int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt = \cos\left(\frac{\pi}{2}x^2\right)$. Therefore $y = 2 + \int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt$ does satisfy the differential equation and the initial condition.

Alternatively, an antiderivative of $\cos\left(\frac{\pi}{2}x^2\right)$ is $y = \int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt + C$. Using the initial condition gives $2 = y(0) = \int_0^0 \cos\left(\frac{\pi}{2}t^2\right) dt + C = 0 + C = C$. Therefore the particular solution to this differential equation and initial condition is $y = \int_0^x \cos\left(\frac{\pi}{2}t^2\right) dt + 2$.

7-7

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = 10 - 2y$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 2$.

(a) Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 0$. Use the tangent line to approximate $f(0.5)$.

(b) Find the value of $\frac{d^2y}{dx^2}$ at the point $(0, 2)$. Is the graph of $y = f(x)$ concave up or concave down at the point $(0, 2)$? Give a reason for your answer.

(c) Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0) = 2$.

(d) For the particular solution $y = f(x)$ found in part (c), find $\lim_{x \to \infty} f(x)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\, y)=(0,\, 2)} = 10 - 2 \cdot 2 = 6$$

An equation for the line tangent to the graph of $y = f(x)$ at $x = 0$ is $y = 6x + 2$.

$f(0.5) \approx 6 \cdot 0.5 + 2 = 5$

7-7

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\left.\dfrac{d^2y}{dx^2}\right|_{(x,\ y)=(0,\ 2)}$
- [ ] concave down with reason

Solution:

$$\frac{d^2y}{dx^2} = -2\frac{dy}{dx} = -2\left(10 - 2y\right) = 4y - 20$$

$$\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(0,\ 2)} = 4 \cdot 2 - 20 = -12$$

Because $\left.\dfrac{d^2y}{dx^2}\right|_{(x,\ y)=(0,\ 2)} < 0$ and $\dfrac{d^2y}{dx^2}$ is continuous, the graph of $y = f\left(x\right)$ is concave down at the point $\left(0,\ 2\right)$.

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

The fourth point requires an expression for y.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- [ ] separation of variables
- [ ] antiderivative
- [ ] constant of integration and uses initial condition

7-7

☐ solves for y

Solution:

$$\frac{1}{10-2y}\,dy = dx$$
$$\int \frac{1}{10-2y}\,dy = \int 1\,dx$$
$$-\frac{1}{2}\ln|10-2y| = x + C$$

Since the solution curve includes the point $(0,\ 2)$,

$$-\frac{1}{2}\ln(10-2y) = x + C$$
$$-\frac{1}{2}\ln(10-2\cdot 2) = 0 + C \ \Rightarrow\ C = -\frac{1}{2}\ln 6$$
$$-\frac{1}{2}\ln(10-2y) = x - \frac{1}{2}\ln 6$$
$$\ln(10-2y) = -2x + \ln 6$$
$$10-2y = e^{-2x+\ln 6} = 6e^{-2x}$$
$$y = 5 - 3e^{-2x}$$

Part D

No supporting work is required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The student response accurately includes a correct value.

Solution:

$$\lim_{x\to\infty}\left(5-3e^{-2x}\right) = 5 - 3\cdot 0 = 5$$

7-7

13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

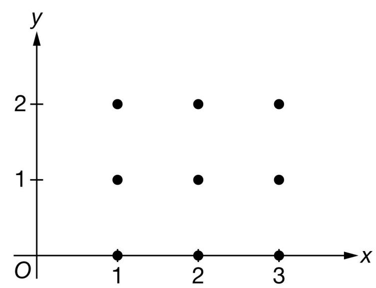
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{y-1}{x}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



(b) Let $y = f(x)$ be the particular solution to the given differential equation with the initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 3$. Use the equation to approximate the value of $f(3.3)$.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(3) = 2$.

Part A

The first point is earned for having all 3 correct segments of zero slope.

The second point is earned for having all 6 correct segments for the other slopes.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

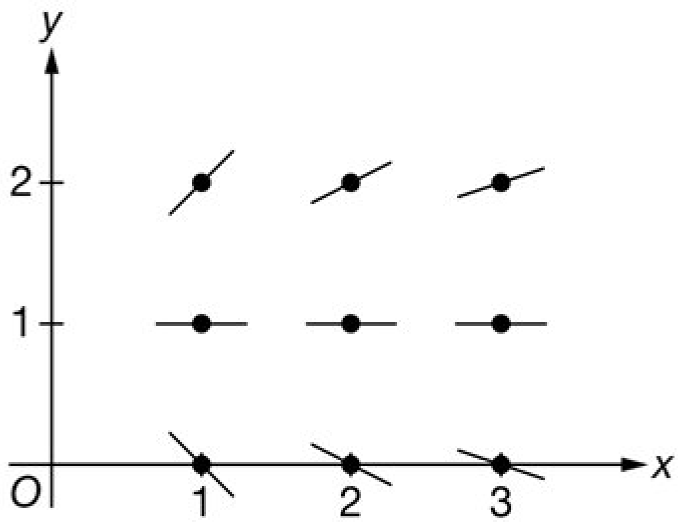
0	1	2

The student response accurately includes both of the criteria below.

7-7

- [ ] zero slopes
- [ ] other slopes

Solution:



Part B

The tangent line equation is not required to be in slope-intercept form.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] tangent line equation
- [ ] approximation

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(3,\ 2)} = \frac{1}{3}$$

An equation for the line tangent to the graph of $y = f\ (x)$ at $x = 3$ is $y = 2 + \frac{1}{3}(x - 3)$.

$f\ (3.3) \approx 2 + \frac{1}{3}(3.3 - 3) = 2.1$

Part C

7-7

Zero out of 5 points earned if no separation of variables.

At most 2 out of 5 points earned [1-1-0-0-0] if no constant of integration.

If response has only one correct antiderivative, the second point is not earned. The response is eligible for the third, fourth, and fifth points if the antiderivative error is carried through the rest of the problem consistently and correctly.

The fifth point requires an expression for y. The domain is included with the solution; this is not a requirement to earn the fifth point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4	5

The student response accurately includes all five of the criteria below.

- [ ] separation of variables
- [ ] both antiderivatives
- [ ] constant of integration
- [ ] uses initial condition
- [ ] solves for y

Solution:

$$\frac{1}{y-1}\,dy = \frac{1}{x}\,dx$$
$$\int \frac{1}{y-1}\,dy = \int \frac{1}{x}\,dx$$
$$\ln|y-1| = \ln|x| + C$$

Since the solution curve includes the point $(3,\ 2)$,

$$\ln(y-1) = \ln x + C$$
$$\ln 1 = \ln 3 + C \ \Rightarrow\ C = -\ln 3$$
$$\ln(y-1) = \ln x - \ln 3 = \ln\left(\tfrac{x}{3}\right)$$
$$y - 1 = \tfrac{x}{3} \ \Rightarrow\ y = 1 + \tfrac{x}{3}$$

Note: This solution is valid for $x > 0$.

7-7

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

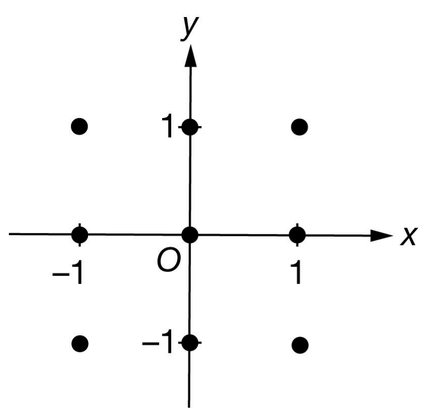
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = xy^3$.

(a) On the axes provided, sketch a slope field for the given differential equation at the 9 points indicated.



(b) Find $\frac{d^2y}{dx^2}$ in terms of x and y. Determine the concavity of all solution curves for the given differential equation in Quadrant IV. Give a reason for your answer.

(c) Find the particular solution $y = f(x)$ to the given differential equation with initial condition $f(2) = -1$.

Part A

The first point is earned for having all 5 correct segments of zero slope.

The second point is earned for having all 4 correct segments for the other slopes.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

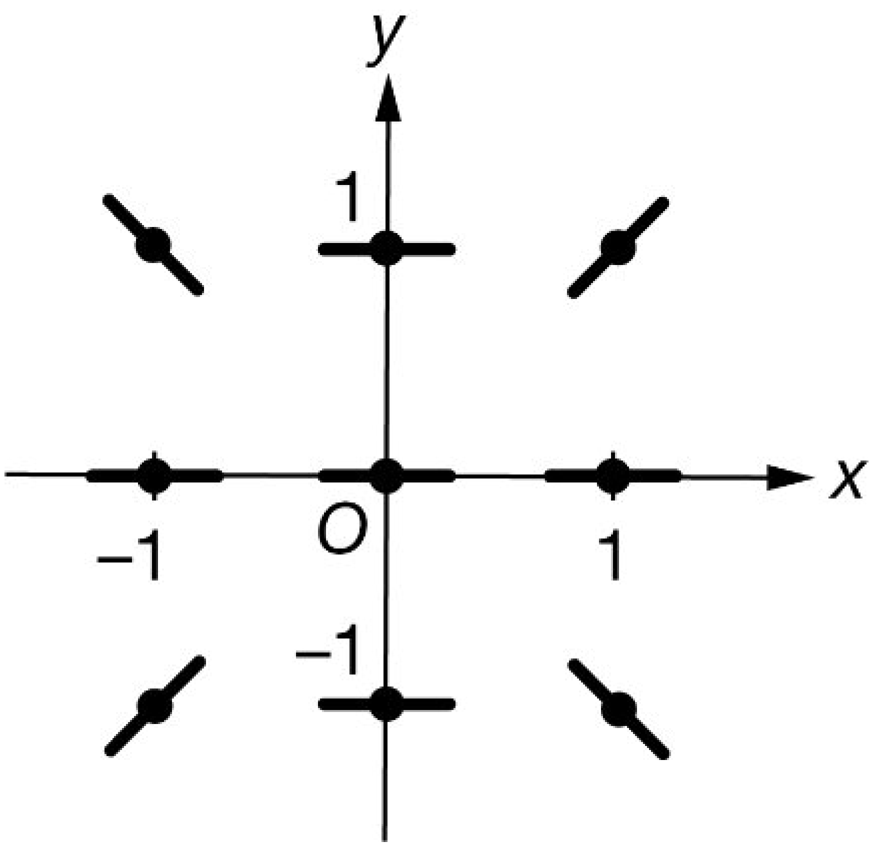
0	1	2

The student response accurately includes both of the criteria below.

7-7

- [ ] zero slopes
- [ ] other slopes

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\frac{d^2y}{dx^2}$
- [ ] concave down with reason

Solution:

$$\frac{d^2y}{dx^2} = y^3 + 3xy^2\frac{dy}{dx} = y^3 + 3xy^2 \cdot xy^3 = y^3 + 3x^2y^5$$

7-7

In Quadrant IV, $x > 0$ and $y < 0$, so $y^3 + 3x^2y^5 < 0$.

Therefore, in Quadrant IV all solution curves are concave down

Part C

Zero out of 5 points earned if no separation of variables.

At most 3 out of 5 points earned [1-1-1-0-0] if no constant of integration.

The fifth point requires an expression for y.

The domain is included with the solution; this is not a requirement to earn the fifth point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4	5

The student response accurately includes all five of the criteria below.

- [ ] separation of variables
- [ ] dy antiderivative
- [ ] dx antiderivative
- [ ] constant of integration and uses initial condition
- [ ] solves for y

Solution:

$$\frac{1}{y^3}\ dy = x\ dx$$

$$\int \frac{1}{y^3}\ dy = \int x\ dx$$

$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$

$$-\frac{1}{2(-1)^2} = \frac{2^2}{2} + C \ \Rightarrow\ C = -\frac{5}{2}$$

$$-\frac{1}{2y^2} = \frac{x^2}{2} - \frac{5}{2} = \frac{x^2-5}{2}$$

$$y^2 = \frac{1}{5-x^2}$$

$y = -\sqrt{\frac{1}{5-x^2}}$, since the solution curve includes the point $(2,\ -1)$.

7-7

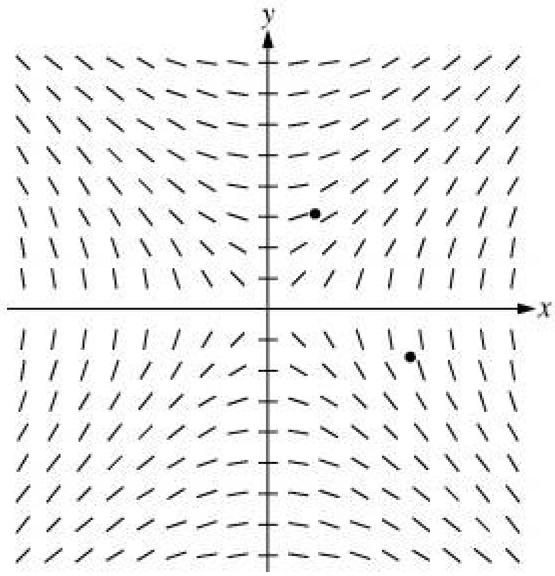
Note: This solution is valid for $-\sqrt{5} < x < \sqrt{5}$.

7-7

15. Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$, where $y \neq 0$.

(a) The slope field for the given differential equation is shown below. Sketch the solution curve that passes through the point (3, -1) and sketch the solution curve that passes through the point (1, 2).

(Note: The points (3, -1) and (1, 2) are indicated in the figure.)



(b) Write an equation for the line tangent to the solution curve that passes through the point (1,2).

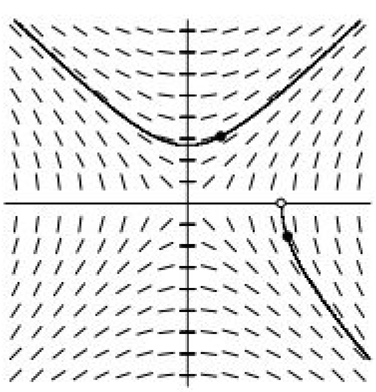
(c) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(3) = -1$ and state its domain.

Part A

1 point is earned for the solution curve through (3, -1)

1 point is earned for the solution curve through (1, 2)

Curves must go through the indicated points, follow the given slope lines, and extend to the boundary of the slope field or the x-axis.



7-7

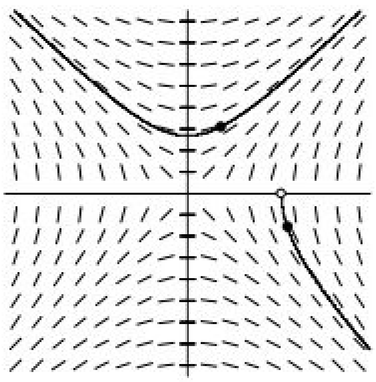
0	1	2

The student response earns all of the following points:

1 point is earned for the solution curve through (3, -1)

1 point is earned for the solution curve through (1, 2)

Curves must go through the indicated points, follow the given slope lines, and extend to the boundary of the slope field or the x-axis.



Part B

1 point is earned for the equation of tangent line

$$\left.\frac{dy}{dx}\right|_{(1,2)} = \frac{1}{2}$$

An equation for the line tangent to the solution curve is $y - 2 = \frac{1}{2}(x - 1)$.

0	1

The student response earns one of the following points:

1 point is earned for the equation of tangent line

7-7

$$\left.\frac{dy}{dx}\right|_{(1,2)} = \frac{1}{2}$$

An equation for the line tangent to the solution curve is $y - 2 = \frac{1}{2}(x - 1)$.

Part C

1 point is earned for separate variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

1 point is earned for domain

$y\,dy = x\,dx$
$\frac{1}{2}y^2 = \frac{1}{2}x^2 + A$
$y^2 = x^2 + C$
$C = -8$

Since the particular solution goes through (3, -1), y must be negative.

$y = -\sqrt{x^2 - 8}$ for $x > \sqrt{8}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for separate variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

7-7

1 point is earned for using initial condition

1 point is earned for solving for y

1 point is earned for domain

$y\,dy = x\,dx$
$\frac{1}{2}y^2 = \frac{1}{2}x^2 + A$
$y^2 = x^2 + C$
$C = -8$

Since the particular solution goes through (3, -1), y must be negative.

$y = -\sqrt{x^2 - 8}$ for $x > \sqrt{8}$

7-7

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

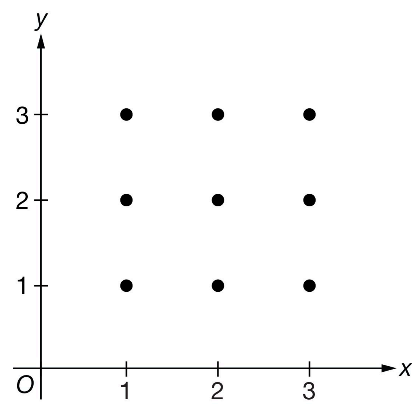
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{2-y}{x}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



(b) Let $y = f(x)$ be the particular solution to the given differential equation with the initial condition $f(3) = 1$. Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 3$. Use the equation to approximate the value of $f(3.5)$.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(3) = 1$.

Part A

The first point is earned for having all 3 correct segments of zero slope.

The second point is earned for having all 6 correct segments for the other slopes.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

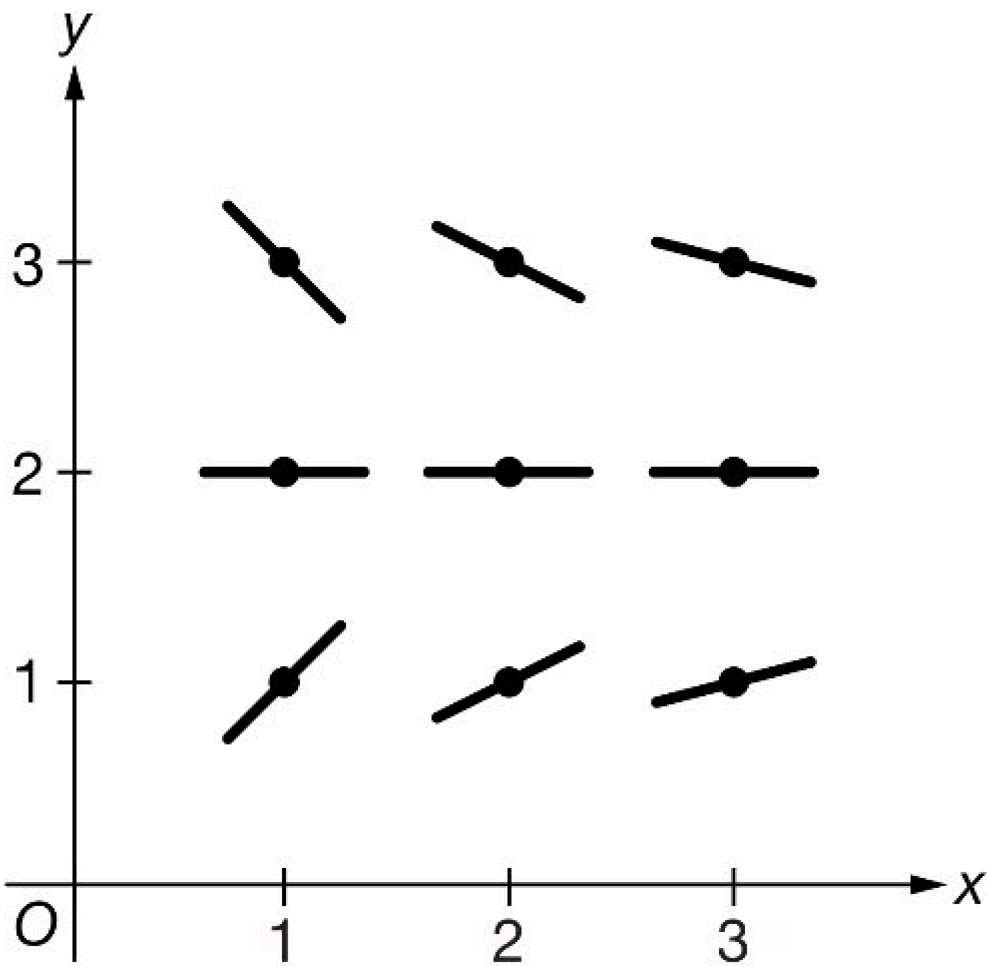
0	1	2

7-7

The student response accurately includes both of the criteria below.

- ☐ zero slopes
- ☐ other slopes

Solution:



Part B

The tangent line equation is not required to be in slope-intercept form.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

7-7

- [ ] tangent line equation
- [ ] approximation

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(3,\ 1)} = \frac{1}{3}$$

An equation for the line tangent to the graph of$y = f\ (x)$ at $x = 3$ is $y = \frac{1}{3}(x - 3) + 1 = \frac{1}{3}x$.

$f\ (3.5) \approx \frac{1}{3}(3.5) = \frac{7}{6}$

Part C

Zero out of 5 points earned if no separation of variables.

At most 2 out of 5 points earned [1-1-0-0-0] if no constant of integration.

The fifth point requires an expression for y. The domain is included with the solution; this is not a requirement to earn the fifth point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4	5

The student response accurately includes all five of the criteria below.

- [ ] separation of variables
- [ ] both antiderivatives
- [ ] constant of integration
- [ ] uses initial condition
- [ ] solves for y

Solution:

$$\frac{1}{2-y}\ dy = \frac{1}{x}\ dx$$

$$\int \frac{1}{2-y}\ dy = \int \frac{1}{x}\ dx$$

$$-\ln|2-y| = \ln|x| + C$$

Since the solution curve includes the point $(3,\ 1)$,

7-7

$$-\ln(2-y) = \ln x + C$$
$$-\ln 1 = \ln 3 + C \;\Rightarrow\; C = -\ln 3$$
$$\ln(2-y) = -\ln x + \ln 3 = \ln\left(\frac{3}{x}\right)$$
$$2 - y = \frac{3}{x} \;\Rightarrow\; y = 2 - \frac{3}{x}$$

1Note: This solution is valid for $x > 0$.

7-7

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

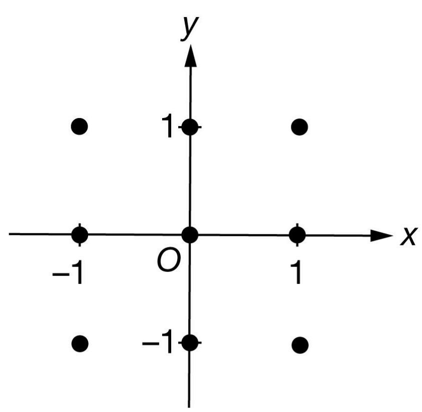
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = xy^4$.

(a) On the axes provided, sketch a slope field for the given differential equation at the 9 points indicated.



(b) Find $\frac{d^2y}{dx^2}$ in terms of x and y. Determine the concavity of all solution curves for the given differential equation in Quadrant II. Give a reason for your answer.

(c) Find the particular solution $y = f(x)$ to the given differential equation with initial condition $f(4) = -1$.

Part A

The first point is earned for having all 5 correct segments of zero slope.

The second point is earned for having all 4 correct segments for the other slopes.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

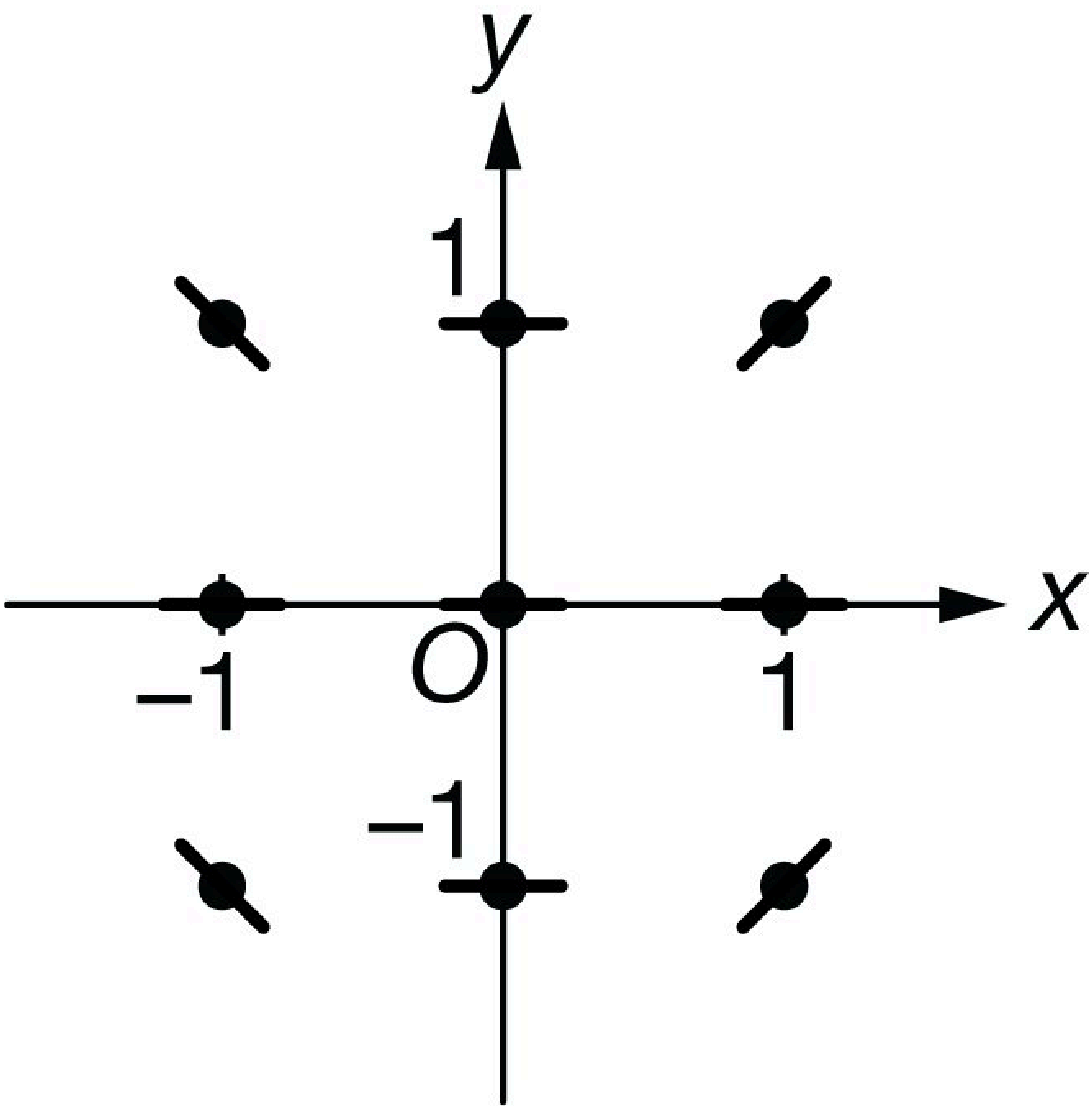
0	1	2

The student response accurately includes both of the criteria below.

7-7

- ☐ zero slopes
- ☐ other slopes

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

7-7

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ $\dfrac{d^2y}{dx^2}$
- ☐ concave up with reason

Solution:

$$\frac{d^2y}{dx^2} = 4xy^3\frac{dy}{dx} + y^4 = 4xy^3 \cdot xy^4 + y^4 = 4x^2y^7 + y^4$$

In Quadrant II, $x < 0$ and $y > 0$, so $4x^2y^7 + y^4 > 0$.

Therefore, in Quadrant II all solution curves are concave up.

Part C

Zero out of 5 points earned if no separation of variables.

At most 3 out of 5 points earned [1-1-1-0-0] if no constant of integration.

The fifth point requires an expression for y. The domain is included with the solution; this is not a requirement to earn the fifth point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4	5 ✓

The student response accurately includes all five of the criteria below.

- ☐ separation of variables
- ☐ dy antiderivative
- ☐ dx antiderivative
- ☐ constant of integration and uses initial condition
- ☐ solves for y

Solution:

7-7

$$\frac{1}{y^4}\ dy = x\ dx$$

$$\int \frac{1}{y^4}\ dy = \int x\ dx$$

$$-\frac{1}{3y^3} = \frac{x^2}{2} + C$$

$$-\frac{1}{3(-1)^3} = \frac{4^2}{2} + C \ \Rightarrow\ C = -\frac{23}{3}$$

$$-\frac{1}{3y^3} = \frac{x^2}{2} - \frac{23}{3} = \frac{3x^2-46}{6}$$

$$y^3 = \frac{2}{46-3x^2}$$

$$y = \sqrt[3]{\frac{2}{46-3x^2}}$$

Note: This solution is valid for $-\sqrt{\frac{46}{3}} < x < \sqrt{\frac{46}{3}}$.

7-7

18. Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$, $where\ y \neq 0$.

Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(3) = -1$ and state its domain.

Part C

1 point is earned for separate variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

1 point is earned for domain

$ydy = xdx$
$\frac{1}{2}y^2 = \frac{1}{2}x^2 + A$
$y^2 = x^2 + C$
$C = -8$

Since the particular solution goes through (3,-1), y must be negative.

$y = -\sqrt{x^2 - 8} for, x > \sqrt{8}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for separate variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

1 point is earned for domain

7-7

$ydy = xdx$
$\frac{1}{2}y^2 = \frac{1}{2}x^2 + A$
$y^2 = x^2 + C$
$C = -8$

Since the particular solution goes through (3,-1), y must be negative.

$y = -\sqrt{x^2 - 8} for, x > \sqrt{8}$

7-7

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

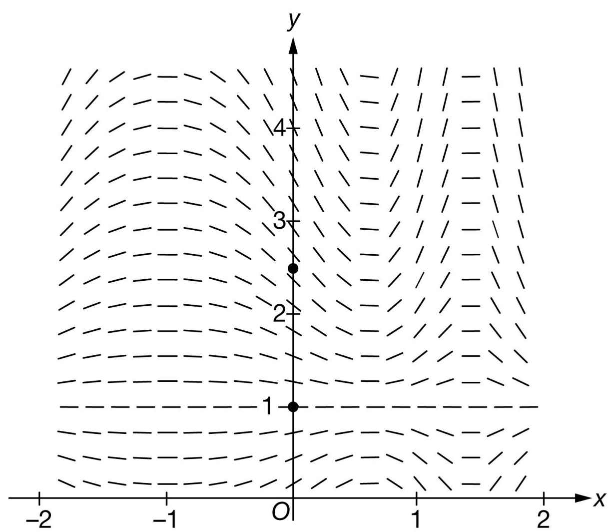
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = (y-1)(x+1)\cos\left(x^2+2x+\pi\right)$.

(a) A portion of the slope field of the differential equation is given below. Sketch the solution curves through the points $(0,1)$ and $\left(0, \frac{5}{2}\right)$.



(b) Write an equation for the line tangent to the solution curve in part (a) at the point $\left(0, \frac{5}{2}\right)$.

(c) Find the particular solution $y = f(x)$ to the differential equation with initial condition $f(0) = \frac{5}{2}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

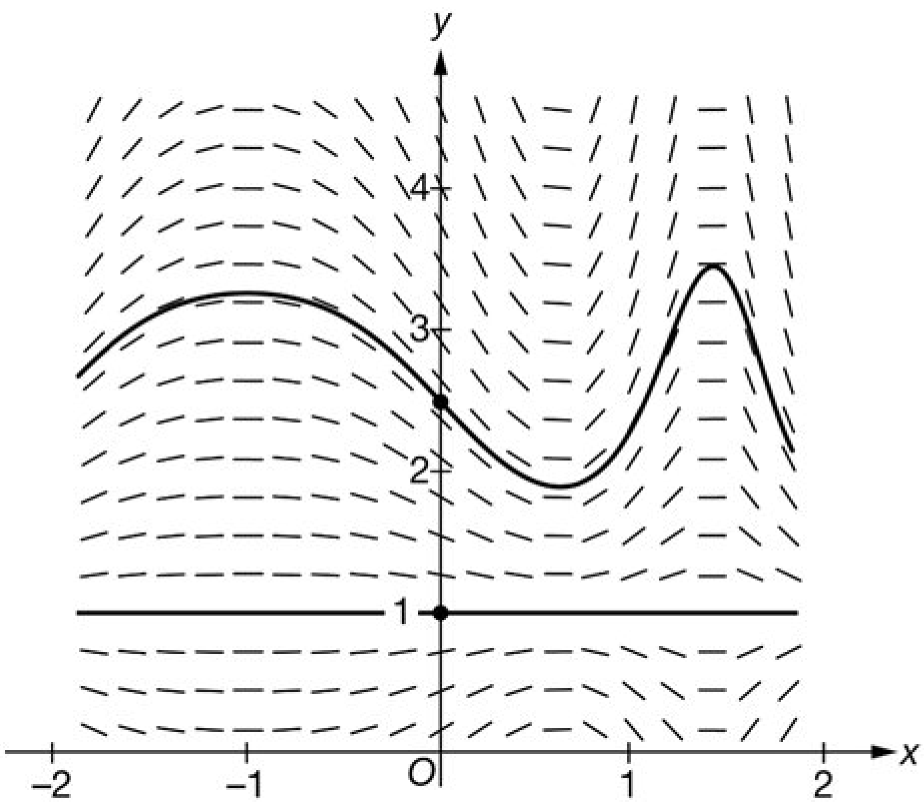
0	1	2 ✓

The student response accurately includes both of the criteria below.

7-7

- ☐ curve through $\left(0,\ \frac{5}{2}\right)$
- ☐ horizontal line through $(0,\ 1)$

Solution:



Part B

The tangent line equation is not required to be in slope-intercept form.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The student response accurately includes a correct tangent line equation.

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=\left(0,\ \frac{5}{2}\right)} = \left(\frac{3}{2}\right)(1)(\cos\ \pi) = -\frac{3}{2}$$

An equation for the tangent line is $y = -\frac{3}{2}x + \frac{5}{2}$.

7-7

Part C

Zero out of 6 points earned if no separation of variables.

At most 4 out of 6 points earned [1-1-1-1-0-0] if no constant of integration.

The sixth point requires an expression for y.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4	5	6

The student response accurately includes all six of the criteria below.

- ☐ separation of variables
- ☐ dy antiderivative
- ☐ substitution
- ☐ dx antiderivative
- ☐ constant of integration and uses initial condition
- ☐ solves for y

Solution:

$$\frac{1}{y-1}\ dy = (x+1)\cos\left(x^2+2x+\pi\right)\ dx$$

$$\int \frac{1}{y-1}\ dy = \int (x+1)\cos\left(x^2+2x+\pi\right)\ dx$$

Let $u = x^2+2x+\pi$. Then $du = (2x+2)\ dx = 2(x+1)\ dx$.

$$\ln|y-1| = \int \frac{1}{2}\cos\ u\ du$$

$$\ln|y-1| = \frac{1}{2}\sin\ u + C = \frac{1}{2}\sin\left(x^2+2x+\pi\right) + C$$

$$\ln\ \frac{3}{2} = \frac{1}{2}\sin\ \pi + C = C$$

$$\ln|y-1| = \frac{1}{2}\sin\left(x^2+2x+\pi\right) + \ln\ \frac{3}{2}$$

$$|y-1| = e^{\frac{1}{2}\sin\left(x^2+2x+\pi\right)+\ln\ \frac{3}{2}} = \frac{3}{2}e^{\frac{1}{2}\sin\left(x^2+2x+\pi\right)}$$

7-7

Because $y(0) = \frac{5}{2}$, $y > 1. \Rightarrow |y-1| = y-1$

$y - 1 = \frac{3}{2} e^{\frac{1}{2}\sin(x^2+2x+\pi)}$

$y = 1 + \frac{3}{2} e^{\frac{1}{2}\sin(x^2+2x+\pi)}$

20. If y = *f*(x) is a solution to the differential equation $\frac{dy}{dx} = e^{x^2}$ with the initial condition *f*(0) = 2, which of the following is true?

(A) $f(x) = 1 + e^{x^2}$

(B) $f(x) = 2xe^{x^2}$

(C) $f(x) = \int_1^x e^{t^2} dt$

(D) $f(x) = 2 + \int_0^x e^{t^2} dt$ ✓

(E) $f(x) = 2 + \int_2^x e^{t^2} dt$

21. Which of the following is the solution to the differential equation $\frac{dy}{dx} = \frac{2y}{2x+1}$ with the initial condition $y(0) = e$ for $x > -\frac{1}{2}$?

(A) $y = e^{2x^2+2x+1}$

(B) $y = 2ex + e$ ✓

(C) $y = \sqrt{x^2 + x + e^2}$

(D) $y = \sqrt{\frac{1}{2}\ln(2x+1) + e^2}$

7-7

22. The following are related to this scenario:

Consider the differential equation $\frac{dy}{dx}=\left(1-\frac{2}{x^2}\right)(y-1)$,where x≠0 . Let *y=f(x)* be the particular solution to the differential equation with initial condition *f*(1)=2 .

Fine the particular solution *y=f(x)* to the differential equation $\frac{dy}{dx}=\left(1-\frac{2}{x^2}\right)(y-1)$ with initial condition *f*(1)=2 .

Part C

The response can earn up to 6 points:

Note: 0/6 if no separation of variables

Note: max 3/6 if no constant of integration.

1 point: For separation of variables

2 point: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of *y*

$$\frac{dy}{dx}=\left(1-\frac{2}{x^2}\right)(y-1)$$

$$\int\frac{dy}{y-1}=\int\left(1-\frac{2}{x^2}\right)dx$$

$$\ln|y-1|=x+\frac{2}{x}+C$$

$$\ln|2-1|=1+\frac{2}{1}+C\Rightarrow C=-3$$

$$\ln|y-1|=x+\frac{2}{x}-3$$

7-7

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

$$\ln(y-1) = x + \frac{2}{x} - 3$$
$$y = f(x) = e^{x+\frac{2}{x}-3} + 1$$

Note: This solution is valid for $x > 0$.

0	1	2	3	4	5	6

The response earns all six of the following points:

1 point: For separation of variables

2 point: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y-1)$$

$$\int \frac{dy}{y-1} = \int \left(1 - \frac{2}{x^2}\right) dx$$

$$\ln |\, y - 1 \,| = x + \frac{2}{x} + C$$

$$\ln |\, 2 - 1 \,| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$

$$\ln |\, y - 1 \,| = x + \frac{2}{x} - 3$$

7-7

Note that $y-1>0$ since the solution curve includes the point $(1,2)$.

$$\ln(y-1)=x+\frac{2}{x}-3$$
$$y=f(x)=e^{x+\frac{2}{x}-3}+1$$

Note: This solution is valid for x>0 .

23. If $f'(x)=\frac{2}{x}$ and $f\left(\sqrt{e}\right)=5$ then $f(e)=$

(A) 2

(B) $\ln 25$

(C) $5+\frac{2}{e}-\frac{2}{e^2}$

(D) 6 ✓

(E) 25

24. Which of the following is the solution to the differential equation $\frac{dy}{dx}=\frac{2xy}{x^2+1}$ whose graph contains the points (0,1)?

(A) $y=e^{x^2}$

(B) $y=x^2+1$ ✓

(C) $y=\ln(x^2+1)$

(D) $y=1+\ln(x^2+1)$

(E) $y=\sqrt{1+2\ln(x^2+1)}$

7-7

25. A student attempts to solve the differential equation $\frac{dy}{dx} = xy^3$ with the initial condition that $y = 2$ when $x = 0$. The steps of the student's solution are shown below.

$$\text{Step 1} : \int \frac{1}{y^3} dy = \int x\, dx$$

$$\text{Step 2} : \ln\left|y^3\right| = \frac{x^2}{2} + C$$

$$\text{Step 3} : \left|y^3\right| = Ke^{x^2/2}$$

$$\text{Step 4} : \left|y^3\right| = 4e^{x^2/2}$$

$$\text{Step 5} : y = \sqrt[3]{4e^{x^2/2}}$$

In which of the following steps does an error first appear?

(A) Step 1

(B) Step 2 ✓

(C) Step 3

(D) Step 4

26. Which of the following is the solution to the differential equation $\frac{dy}{dx} = e^{y+x}$ with the initial condition $y(0) = -\ln 4$?

(A) $y = -x - \ln 4$

(B) $y = x - \ln 4$

(C) $y = -\ln(-e^x + 5)$ ✓

(D) $y = -\ln(e^x + 3)$

(E) $y = \ln(e^x + 3)$

27. What is the particular solution to the differential equation $\frac{dy}{dx} = xy^2$ with the initial condition $y(2) = 1$?

(A) $y = e^{\frac{x^2}{2} - 2}$

(B) $y = e^{\frac{x^2}{2}}$

(C) $y = -\frac{2}{x^2}$

(D) $y = \frac{2}{6 - x^2}$ ✓

(E) $y = \frac{6 - x^2}{2}$

7-7

28. A student attempted to solve the differential equation $\frac{dy}{dx} = xy$ with initial condition $y = 2$ when $x = 0$. In which step, if any, does an error first appear?

Step 1: $\int \frac{1}{y} dy = \int x dx$

Step 2: $\ln |y| = \frac{x^2}{2} + C$

Step 3: $|y| = e^{x^2/2} + C$

Step 4: Since $y = 2$ when $x = 0$, $2 = e^0 + C$.

Step 5: $y = e^{x^2/2} + 1$

(A) Step 2

(B) Step 3 ✓

(C) Step 4

(D) Step 5

(E) There is no error in the solution.

29. Which of the following is the solution to the differential equation $\frac{dy}{dx} = 5y^2$ with the initial condition y(0) = 3?

(A) $y = \sqrt{9e^{5x}}$

(B) $y = \sqrt{\frac{1}{9}e^{5x}}$

(C) $y = \sqrt{e^{5x} + 9}$

(D) $y = \frac{3}{1-15x}$ ✓

(E) $y = \frac{3}{1+15x}$

30. Which of the following is the solution to the differential equation $\frac{dy}{dx} = -2xy$ with the initial condition y(1) = 4?

(A) $y = e^{x^2} + 4 - e$

(B) $y = e^{-x^2} + 4 - \frac{1}{e}$

(C) $y = 4e^{x^2-1}$

(D) $y = 4e^{-x^2+1}$ ✓

(E) $y = e^{-x^2+16}$

7-7

31. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A large vat is initially filled with a saltwater solution. A solution with a higher concentration of salt flows into the vat, and solution flows out of the vat at the same rate. The number of pounds of salt in the vat at time t minutes is modeled by the function A that satisfies the differential equation $\frac{dA}{dt} = 6 - 0.02A$. At time $t = 10$ minutes, the vat contains 50 pounds of salt.

(a) Write an equation for the line tangent to the graph of A at $t = 10$. Use the tangent line to approximate the number of pounds of salt in the vat at time $t = 12$ minutes.

(b) Show that $A(t) = 300 - 250e^{0.2-0.02t}$ satisfies the differential equation $\frac{dA}{dt} = 6 - 0.02A$ with initial condition $A(10) = 50$.

(c) The flow of solution into the vat is stopped, and the solution is drained. The depth of solution in the vat is modeled by the function h that satisfies the differential equation $\frac{dh}{dt} = -k\sqrt{h}$, where $h(t)$ is measured in meters, t is the number of minutes since draining began, and k is a constant. If the depth of the solution is 16 meters at time $t = 0$ minutes and 4 meters at time $t = 30$ minutes, what is $h(t)$ in terms of t ?

Part A

For the second point, it is incorrect to state $A(12) = 60$ rather than $A(12) \approx 60$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

7-7

$A(10) = 50$

$\left.\frac{dA}{dt}\right|_{t=10} = 6 - 0.02 \cdot 50 = 5$

An equation for the line tangent to the graph of A at $t = 10$ is $y = 50 + 5(t - 10)$.

At time $t = 12$ minutes, the vat contains approximately $50 + 5(12 - 10) = 60$ pounds of salt. $A(12) \approx 60$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- [ ] verification of initial condition
- [ ] $\frac{dA}{dt} = 5e^{0.2-0.02t}$
- [ ] verification that $\frac{dA}{dt} = 6 - 0.02A$

Solution:

$A(10) = 300 - 250e^{0.2-0.02\cdot 10} = 300 - 250e^{0} = 50$

$$\frac{dA}{dt} = \frac{d}{dt}\left(300 - 250e^{0.2-0.02t}\right)$$
$$= -250e^{0.2-0.02t}(-0.02) = 5e^{0.2-0.02t}$$

$$6 - 0.02A = 6 - 0.02\left(300 - 250e^{0.2-0.02t}\right)$$
$$= 6 - 6 + 5e^{0.2-0.02t} = 5e^{0.2-0.02t}$$

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

Both antiderivatives must be correct to earn the second point.

The fourth point requires an expression for $h(t)$. The domain of $h(t)$ is included with the solution; this is not a requirement to earn the fourth point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response

7-7

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- ☐ separation of variables
- ☐ antiderivatives
- ☐ constant of integration and uses initial conditions
- ☐ $h\ (t)$

Solution:

$$\frac{dh}{dt} = -k\sqrt{h}$$
$$\frac{dh}{\sqrt{h}} = -k\ dt$$
$$\int \frac{1}{\sqrt{h}}\ dh = -\int k\ dt$$
$$2\sqrt{h} = -kt + C$$
$$2\sqrt{16} = 0 + C \ \Rightarrow \ C = 8$$
$$2\sqrt{4} = -30k + 8 \ \Rightarrow \ k = \frac{2}{15}$$
$$2\sqrt{h} = -\frac{2}{15}t + 8$$
$$h\ (t) = \left(-\frac{t}{15} + 4\right)^2$$

Note: This is valid for $0 \leq t \leq 60$.

7-7

32. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A book in a popular book series goes on sale at time $t = 0$. The total number of books sold by time t hours is modeled by the function B that satisfies the differential equation $\frac{dB}{dt} = \frac{5}{\sqrt{t+4}} + \frac{1}{9}(t-6)^2$ with $B(12) = 36$, where B is measured in hundreds of books and $0 \le t \le 24$.

(a) Write an equation for the line tangent to the graph of B at $t = 12$. Use this equation to approximate the total number of books sold by time $t = 10$.

(b) Find $B''(5)$. Using correct units, interpret the meaning of $B''(5)$ in the context of the problem.

(c) Find an expression for $B(t)$. Based on this model, what is the total number of books sold by time $t = 21$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$B(12) = 36$

$$B'(12) = \frac{5}{\sqrt{12+4}} + \frac{1}{9}(12-6)^2 = \frac{5}{4} + 4 = \frac{21}{4}$$

An equation for the line tangent to the graph of B at $t = 12$ is $y = 36 + \frac{21}{4}(t - 12)$.

7-7

$B\left(10\right) \approx 36 + \frac{21}{4}(10-12) = 36 - \frac{21}{2} = \frac{51}{2} = 25.5$

The number of books sold by time $t=10$ is approximately 2550.

Part B

The third point requires (1) rate of $-\frac{17}{54}$, (2) units of hundred books per hour per hour, and (3) reference to $t=5$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ $B''\left(t\right)$
- ☐ $B''\left(5\right)$
- ☐ interpretation

Solution:

$$B''\left(t\right) = 5\left(-\tfrac{1}{2}\right)(t+4)^{-\frac{3}{2}} + \tfrac{1}{9}\cdot 2\left(t-6\right) = -\frac{5}{2(t+4)^{\frac{3}{2}}} + \tfrac{2}{9}(t-6)$$

$$B''\left(5\right) = -\frac{5}{2(5+4)^{\frac{3}{2}}} + \tfrac{2}{9}(5-6) = -\frac{5}{54} - \frac{2}{9} = -\frac{17}{54}$$

At time $t=5$, the rate at which books are selling is changing at a rate of $-\frac{17}{54}$ hundred books per hour per hour.

Part C

At most 1 out of 4 points earned [1-0-0-0] if no constant of integration.

The fourth point requires both an expression for $B\left(t\right)$ and value for $B\left(21\right)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

7-7

- ☐ antiderivative
- ☐ constant of integration
- ☐ uses initial conditions
- ☐ $B\left(t\right)$ and $B\left(21\right)$

Solution:

$$\frac{dB}{dt} = \frac{5}{\sqrt{t+4}} + \frac{1}{9}(t-6)^2$$

$$B\left(t\right) = 5 \cdot 2\sqrt{t+4} + \frac{1}{9} \cdot \frac{1}{3}(t-6)^3 + C = 10\sqrt{t+4} + \frac{1}{27}(t-6)^3 + C$$

$$B\left(12\right) = 10\sqrt{12+4} + \frac{1}{27}(12-6)^3 + C = 48 + C = 36 \;\Rightarrow\; C = -12$$

$$B\left(t\right) = 10\sqrt{t+4} + \frac{1}{27}(t-6)^3 - 12$$

$$B\left(21\right) = 10\sqrt{21+4} + \frac{1}{27}(21-6)^3 - 12 = 50 + 125 - 12 = 163$$

The number of books sold by time $t = 21$ is 16,300.

7-7

33. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The population of a virus in a host can be modeled by the function A that satisfies the differential equation $\frac{dA}{dt} = t - \frac{1}{5}A$, where A is measured in millions of virus cells and t is measured in days for $5 \leq t < 10$. At time $t = 5$ days, there are 10 million cells of the virus in the host.

(a) Write an equation for the line tangent to the graph of A at $t = 5$. Use the tangent line to approximate the number of virus cells in the host, in millions, at time $t = 7$ days.

(b) Show that $A(t) = -25 + 5t + 10e^{-\frac{t}{5}+1}$ satisfies the differential equation $\frac{dA}{dt} = t - \frac{1}{5}A$ with initial condition $A(5) = 10$.

(c) The host receives an antiviral medication. The amount of medication in the host is modeled by the function M that satisfies the differential equation $\frac{dM}{dt} = -\frac{1}{2}\left(\frac{M}{t+k}\right)$, where M is measured in milligrams, t is measured in days since the host received the medication, and k is a positive constant. If the amount of medication in the host is 30 milligrams at time $t = 0$ days and 15 milligrams at time $t = 3$ days, what is $M(t)$ in terms of t ?

Part A

For the second point, it is incorrect to state $A(7) = 16$ rather than $A(7) \approx 16$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$A(5) = 10$

7-7

$$\left.\frac{dA}{dt}\right|_{t=5} = 5 - \tfrac{1}{5} \cdot 10 = 3$$

An equation for the line tangent to the graph of A at $t = 5$ is $y = 10 + 3\,(t - 5)$.

At time $t = 7$ days, there are approximately $10 + 3\,(7 - 5) = 16$ million virus cells in the host. $A\,(7) \approx 16$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ verification of initial condition
- ☐ $\frac{dA}{dt} = 5 - 2e^{-\frac{t}{5}+1}$
- ☐ verification that $\frac{dA}{dt} = t - \frac{1}{5}A$

Solution:

$$A\,(5) = -25 + 5 \cdot 5 + 10e^{-1+1} = 10$$

$$\frac{dA}{dt} = \frac{d}{dt}\left(-25 + 5t + 10e^{-\frac{t}{5}+1}\right) = 5 + 10e^{-\frac{t}{5}+1}\left(-\tfrac{1}{5}\right) = 5 - 2e^{-\frac{t}{5}+1}$$

$$t - \tfrac{1}{5}A = t - \tfrac{1}{5}\left(-25 + 5t + 10e^{-\frac{t}{5}+1}\right) = t + 5 - t - 2e^{-\frac{t}{5}+1} = 5 - 2e^{-\frac{t}{5}+1}$$

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

Both antiderivatives must be correct to earn the second point.

The fourth point requires an expression for $M\,(t)$. The domain of $M\,(t)$ is included with the solution; this is not a requirement to earn the fourth point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

7-7

The student response accurately includes all four of the criteria below.

- [ ] separation of variables
- [ ] antiderivatives
- [ ] constant of integration and uses initial conditions
- [ ] $M(t)$

Solution:

$$\frac{dM}{dt} = -\frac{1}{2}\left(\frac{M}{t+k}\right)$$
$$\frac{dM}{M} = -\frac{1}{2}\left(\frac{dt}{t+k}\right)$$
$$\int \frac{1}{M}\, dM = -\frac{1}{2}\int \left(\frac{1}{t+k}\right)\, dt$$
$$\ln|M| = -\frac{1}{2}\ln|t+k| + C$$

From the context, M and t are nonnegative. It is given that $k > 0$.

$$\ln M = -\frac{1}{2}\ln(t+k) + C$$
$$\ln M = \ln\frac{1}{\sqrt{t+k}} + C$$
$$M = \frac{D}{\sqrt{t+k}}$$
$$30 = \frac{D}{\sqrt{k}} \Rightarrow D = 30\sqrt{k}$$
$$15 = \frac{D}{\sqrt{3+k}} \Rightarrow D = 15\sqrt{3+k}$$
$$30\sqrt{k} = 15\sqrt{3+k} \Rightarrow k = 1 \text{ and } D = 30$$
$$M(t) = \frac{30}{\sqrt{t+1}}$$

Note: This is valid for $t \geq 0$.

7-7

34. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

On a certain day at a diner, the total number of food orders taken by time t is modeled by the function F that satisfies the differential equation $\frac{dF}{dt} = \frac{48}{\sqrt{3t+1}}$ with $F(1) = 32$, where t is the number of hours since the diner opened and $0 \le t \le 8$.

(a) Write an equation for the line tangent to the graph of F at $t = 1$. Use this equation to approximate the total number of food orders taken by time $t = 2$.

(b) Find $F''(5)$. Using correct units, interpret the meaning of $F''(5)$ in the context of the problem.

(c) Find an expression for $F(t)$. Based on this model, what is the total number of food orders taken by time $t = 8$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$F(1) = 32$

$$F'(1) = \frac{48}{\sqrt{3 \cdot 1+1}} = 24$$

An equation for the line tangent to the graph of F at $t = 1$ is $y = 32 + 24(t - 1)$.

By time $t = 2$, the total number of food orders taken is approximately $32 + 24(2 - 1) = 54$.

7-7

Part B

The third point requires (1) rate of $-\frac{9}{8}$, (2) units of orders per hour per hour, and (3) reference to $t = 5$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] $F''(t)$
- [ ] $F''(5)$
- [ ] interpretation

Solution:

$$F''(t) = 48\left(-\frac{1}{2}\right)(3t+1)^{-\frac{3}{2}} \cdot 3 = -72(3t+1)^{-\frac{3}{2}}$$

$$F''(5) = -72(3 \cdot 5 + 1)^{-\frac{3}{2}} = -72(16)^{-\frac{3}{2}} = -\frac{72}{64} = -\frac{9}{8}$$

At time $t = 5$, the rate at which food orders are taken is changing at a rate of $-\frac{9}{8}$ orders per hour per hour.

Part C

At most 1 out of 4 points earned [1-0-0-0] if no constant of integration.

The fourth point requires both an expression for $F(t)$ and value for $F(8)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- [ ] antiderivative
- [ ] constant of integration
- [ ] uses initial conditions

7-7

- [ ] $F(t)$ and $F(8)$

Solution:

$\frac{dF}{dt} = \frac{48}{\sqrt{3t+1}}$

$F(t) = 48 \cdot 2\sqrt{3t+1} \cdot \frac{1}{3} + C = 32\sqrt{3t+1} + C$

$F(1) = 32\sqrt{3 \cdot 1 + 1} + C = 64 + C = 32 \;\Rightarrow\; C = -32$

$F(t) = 32\sqrt{3t+1} - 32$

The number of food orders taken by time $t = 8$ is $32\sqrt{3 \cdot 8 + 1} - 32 = 128$.

7-7

35. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

At an aquarium, a tank is kept full of water. At time $t = 0$, where t is measured in minutes, the water in the tank contains no salt. For time $t > 0$, water containing salt flows into the tank at a constant rate and mixes with the water in the tank. The mixed water leaves the tank through an overflow drain at the same constant rate. The number of pounds of salt in the water in the tank at time $t \geq 0$ is modeled by the function S, which satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{6}(S - 36)$ with initial condition $S(0) = 0$.

(a) Is the amount of salt in the water in the tank changing more rapidly when the water contains 6 pounds of salt or when the water contains 30 pounds of salt? Explain your reasoning.

(b) Write an expression for $\frac{d^2S}{dt^2}$ in terms of S. Find the value of $\frac{d^2S}{dt^2}$ at the instant when there are 20 pounds of salt in the water in the tank. Using correct units, interpret this value in the context of the problem.

(c) Use separation of variables to find $y = S(t)$, the particular solution to the differential equation $\frac{dS}{dt} = -\frac{1}{6}(S - 36)$ with initial condition $S(0) = 0$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an attempt to evaluate $\dfrac{dS}{dt}$ at either $S = 6$ or $S = 30$.

Correct, labeled values of $\dfrac{dS}{dt}$ at both $S = 6$ and $S = 30$ are required to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Uses $\dfrac{dS}{dt}$
- ☐ Answer with reason

7-7

Solution:

$$\left.\frac{dS}{dt}\right|_{S=6} = -\tfrac{1}{6}(6 - 36) = 5$$

$$\left.\frac{dS}{dt}\right|_{S=30} = -\tfrac{1}{6}(30 - 36) = 1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must present $\frac{d^2S}{dt^2}$ in terms of S in order to earn the first point.

A response must present a negative value of $\left.\frac{d^2S}{dt^2}\right|_{S=20}$ in order to be eligible for the third point.

A response with a sign error in the second derivative (i.e., $\frac{d^2S}{dt^2} = -\frac{1}{36}(S - 36)$) can earn the second point for calculating $\left.\frac{d^2S}{dt^2}\right|_{S=20} = -\frac{1}{36}(20 - 36) = \frac{4}{9}$. This response is not eligible for the third point and thus earns a maximum of 1 point in part (b).

The interpretation requires "at the moment when there are 20 pounds of salt," "rate of the amount of salt," "decreasing," and "at a rate of $|\text{the presented value}|$ pounds per minute per minute" (or the equivalent) to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $\frac{d^2S}{dt^2}$
- ☐ $\left.\frac{d^2S}{dt^2}\right|_{S=20}$
- ☐ Interpretation with units

Solution:

$$\frac{d^2S}{dt^2} = -\tfrac{1}{6}\frac{dS}{dt} = -\tfrac{1}{6}\left(-\tfrac{1}{6}\right)(S - 36)$$

$$\left.\frac{d^2S}{dt^2}\right|_{S=20} = -\tfrac{1}{6}\left(-\tfrac{1}{6}\right)(20 - 36) = -\tfrac{4}{9}$$

At the instant when there are 20 pounds of salt in the tank, the rate of change of the number of pounds of salt is decreasing at a rate of $\frac{4}{9}$ pound per minute per minute.

7-7

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 4 points.

A response of $-\frac{1}{6}\int \frac{1}{S-36}\, dS = \int dt$ is a bad separation and does not earn the first point. However, this response is eligible for the second and third points. It cannot earn the fourth point.

A response that enters with both correct antiderivatives earns the first 2 points.

If the absolute value is missing on the antiderivative of $\frac{1}{S-36}$ the response earns the second point and is eligible for the third point. The response is not eligible for the fourth point.

A response with no constant of integration can earn at most the first 2 points.

A response is eligible for the third point if it has earned the first point and has at least one of the two antiderivatives correct. A response earns the third point by correctly including the constant of integration in an equation involving the presented antiderivatives and then replacing t with 0 and S with 0. This point is earned even in the presence of nonreal values (e.g., $\ln(-36)$) or severe algebraic errors.

A response is eligible for the fourth point only if it has earned the first 3 points and has correctly handled absolute value in all steps.

There are alternate (equivalent) algebraic forms of the correct equation, including, but not limited to:
$S(t) = 36 - e^{\ln 36 - t/6}$ and $S(t) = 36 - \frac{36}{e^{t/6}}$.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- ☐ Separation of variables
- ☐ Antiderivatives
- ☐ Constant of integration and uses initial condition
- ☐ Solves for S

Solution:

$$\frac{dS}{dt} = -\frac{1}{6}(S-36)$$

7-7

$$\int \frac{1}{S-36}\, dS = \int -\frac{1}{6}\, dt$$

$\ln|S-36| = -\frac{1}{6}t + C$

Because $0 \le S < 36$, $|S-36| = 36 - S$.

$\ln(36-0) = -\frac{1}{6}(0) + C \;\Rightarrow\; \ln(36) = C$

$36 - S = 36e^{-t/6}$

$S(t) = 36 - 36e^{-t/6},\; t \ge 0$

7-7

36. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

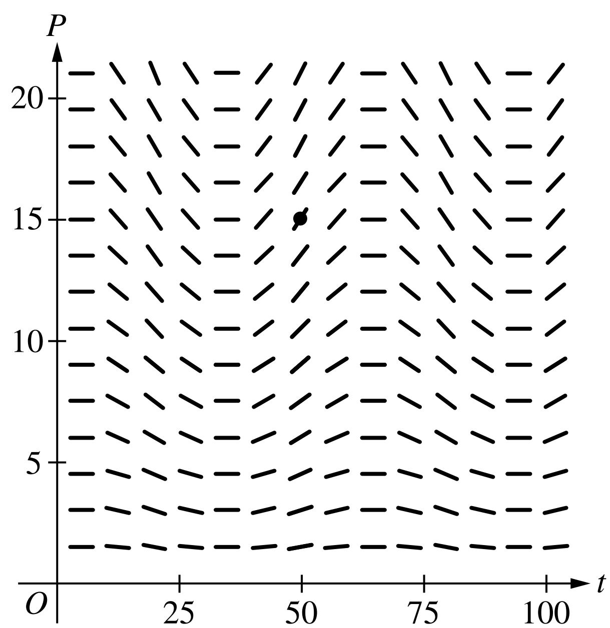
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Scientists study a population that experiences variations in its rate of growth. The rate of growth of the population is modeled by the differential equation $\frac{dP}{dt} = P \cdot 0.02 \cos\left(\frac{\pi}{30}(t+10)\right)$, where t is measured in days since the observation of the population began. Let $P = f(t)$ be the particular solution to the given differential equation satisfying $f(50) = 15$.

(a) A slope field for the given differential equation is shown. Sketch the solution curve that passes through the point $(50, 15)$.



(b) Write an equation for the line tangent to the graph of $P = f(t)$ at time $t = 50$.

(c) It is known that $f'(t) > 0$ and $f''(t) > 0$ on the interval $47 < t < 51$. If the tangent line found in part (b) is used to approximate $f(48)$, is the approximation an underestimate or an overestimate for the actual value of $f(48)$? Give a reason for your answer.

(d) Use separation of variables to find the particular solution $P = f(t)$ to the differential equation $\frac{dP}{dt} = P \cdot 0.02 \cos\left(\frac{\pi}{30}(t+10)\right)$ that satisfies $f(50) = 15$.

Part A

7-7

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $(50,\ 15)$, extend reasonably close to the left and right edges of the slope field, and have no obvious conflicts with the given line segments.

Only portions of the solution curve within the given slope field are considered.

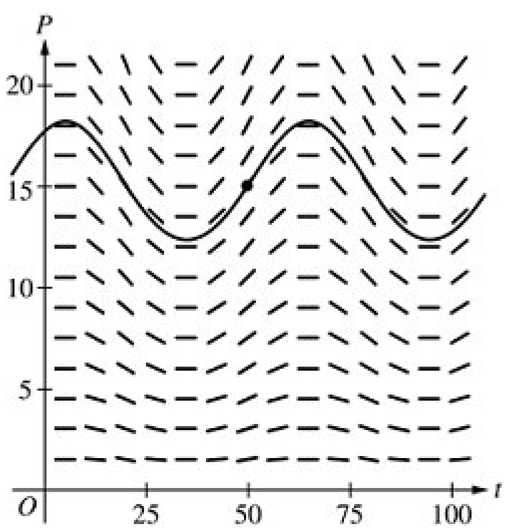
All local maximum/minimum points on the solution curve must occur at horizontal line segments in the slope field.

0	1

The student response accurately includes the criteria below.

- [ ] Solution curve through $(50,\ 15)$,

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form and can be written in terms of any variables.

The second point can be earned with a tangent line through the point $(50,\ 15)$ with a slope equal to the declared value of $\left.\frac{dP}{dt}\right|_{(50,\ 15)}$.

0	1	2

7-7

The student response accurately includes **both** of the criteria below.

- [ ] Slope at $(50, 15)$
- [ ] Tangent line equation

Solution:

$f(50) = 15$

$f'(50) = \left.\frac{dP}{dt}\right|_{(50,15)} = 15 \cdot 0.02\cos\left(\frac{\pi}{30} \cdot 60\right) = 0.3$

An equation for the line tangent to the graph of $P = f(t)$ at time $t = 50$ is $y = 15 + 0.3(t - 50)$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The reason must include one of $f''(t) > 0$, $f'(t)$ is increasing, or $f(t)$ is concave up.

The point is not earned if the reason includes statements about the behavior of $f'(t)$ or $f''(t)$ outside the interval $47 < t < 51$.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Underestimate with reason

Solution:

The tangent line approximation of $f(48)$ is an underestimate because $f''(t) > 0$ on an interval that contains $t = 48$ and $t = 50$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 5 points.

A response that presents an antiderivative of without absolute value symbols is sufficient for the second point.

A response with no constant of integration can earn at most the first 3 points.

7-7

A response is eligible for the fourth point only if it has earned the first point and at least 1 of the 2 antiderivative points.

An eligible response earns the fourth point by correctly including the constant of integration in an equation and substituting 50 for t and P. A response is eligible for the fifth point only if it has earned the first 4 points.

0	1	2	3	4	5 ✓

The student response accurately includes **all five** of the criteria below.

- ☐ Separates variables
- ☐ Finds one antiderivative
- ☐ Finds the other antiderivative
- ☐ Constant of integration and uses initial condition
- ☐ Solves for P

Solution:

$$\int \frac{dP}{P} = \int 0.02\cos\left(\frac{\pi}{30}(t+10)\right)\, dt$$

$$\ln|P| = \frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(t+10)\right) + C$$

$$\ln|15| = \frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(50+10)\right) + C \Rightarrow C = \ln 15$$

Because $P(50) > 0$, $\ln|P| = \ln P$.

$$\ln P = \frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(t+10)\right) + \ln 15$$

$$P = 15e^{\frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(t+10)\right)}$$

7-7

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

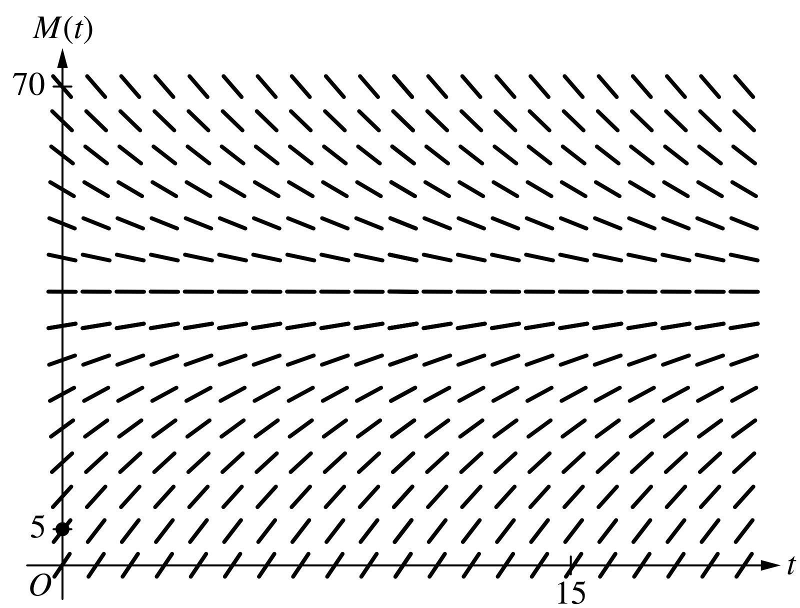
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A bottle of milk is taken out of a refrigerator and placed in a pan of hot water to be warmed. The increasing function M models the temperature of the milk at time t, where $M(t)$ is measured in degrees Celsius ($^\circ$C) and t is the number of minutes since the bottle was placed in the pan. M satisfies the differential equation $\frac{dM}{dt} = \frac{1}{4}(40 - M)$. At time $t = 0$, the temperature of the milk is 5°C. It can be shown that $M(t) < 40$ for all values of t.

(a) A slope field for the differential equation $\dfrac{dM}{dt} = \dfrac{1}{4}(40 - M)$ is shown. Sketch the solution curve through the point $(0, 5)$.



(b) Use the line tangent to the graph of M at $t = 0$ to approximate $M(2)$, the temperature of the milk at time $t = 2$ minutes.

(c) Write an expression for $\dfrac{d^2M}{dt^2}$ in terms of M. Use $\dfrac{d^2M}{dt^2}$ to determine whether the approximation from part (b) is an underestimate or an overestimate for the actual value of $M(2)$. Give a reason for your answer.

7-7

(d) Use separation of variables to find an expression for $M(t)$, the particular solution to the differential equation $\frac{dM}{dt} = \frac{1}{4}(40 - M)$ with initial condition $M(0) = 5$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $(0, 5)$, extend reasonably close to the left and right edges of the rectangle, and have no obvious conflicts with the given slope lines.

Only portions of the solution curve within the given slope field are considered.

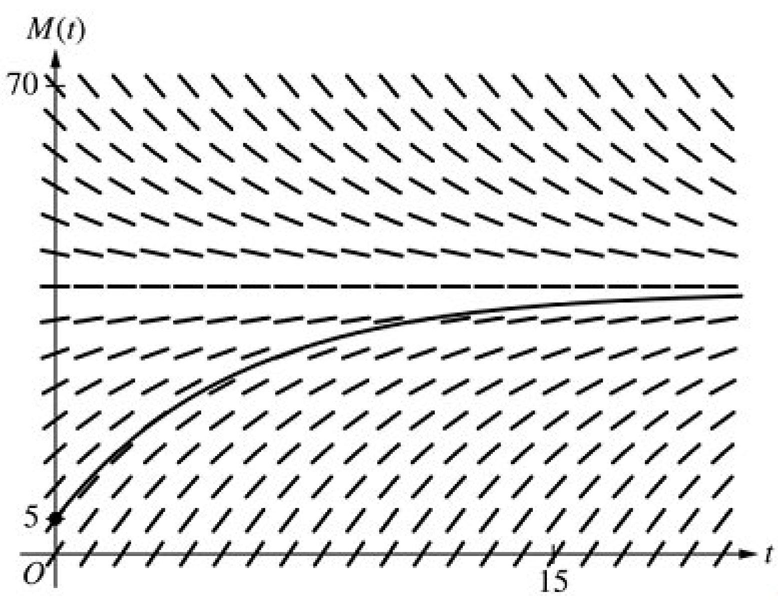
The solution curve must lie entirely below the horizontal line segments at $M = 40$.

0	1

The student response accurately includes the criteria below.

☐ Solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The value of the slope may appear in a tangent line equation or approximation.

7-7

A response of $5+\frac{35}{4}\cdot 2$ is the minimal response to earn both points.

A response of $\frac{1}{4}(40-5)$ earns the first point. If there are any subsequent errors in simplification, the response does not earn the second point.

In order to earn the second point the response must present an approximation found by using a tangent line that:

- passes through the point $(0,5)$ and
- has slope $\frac{35}{4}$ or a nonzero slope that is declared to be the value of $\frac{dM}{dt}$.

An unsupported approximation does not earn the second point.

The approximation need not be simplified, but the response does not earn the second point if the approximation is simplified incorrectly.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\left.\frac{dM}{dt}\right|_{t=0}$
- ☐ Approximation

Solution:

$$\left.\frac{dM}{dt}\right|_{t=0}=\frac{1}{4}(40-5)=\frac{35}{4}$$

The tangent line equation is $y=5+\frac{35}{4}(t-0)$.

$$M(2)\approx 5+\frac{35}{4}\cdot 2=22.5$$

The temperature of the milk at time $t=2$ minutes is approximately 22.5° Celsius.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $\frac{d^2M}{dt^2}=-\frac{1}{4}\left(\frac{1}{4}(40-M)\right)$ or $\frac{d^2M}{dt^2}=-\frac{1}{16}(40-M)$ (or equivalent). A response that presents any subsequent simplification error does not earn the second point.

7-7

A response that presents an expression for $\frac{d^2M}{dt^2}$ in terms of $\frac{dM}{dt}$ but fails to continue to an expression in terms of (i.e., $\frac{d^2M}{dt^2} = -\frac{1}{4}\frac{dM}{dt}$) does not earn the first point but is eligible for the second point.

If the response presents an expression for $\frac{d^2M}{dt^2}$ that is incorrect, the response is eligible for the second point only if the expression is a nonconstant linear function that is negative for $5 < M < 40$.

- Special case: A response that presents $\frac{d^2M}{dt^2} = \frac{1}{16}(40 - M)$ does not earn the first point but is eligible to earn the second point for a consistent answer and reason.

To earn the second point a response must include $\frac{d^2M}{dt^2} < 0$, or $\frac{dM}{dt}$ is decreasing, or the graph of M is concave down, as well as the conclusion that the approximation is an overestimate.

A response that presents an argument based on $\frac{d^2M}{dt^2}$ or concavity at a single point does not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{d^2M}{dt^2}$
- ☐ Overestimate with reason

Solution:

$$\frac{d^2M}{dt^2} = -\frac{1}{4}\frac{dM}{dt} = -\frac{1}{4}\left(\frac{1}{4}(40 - M)\right) = -\frac{1}{16}(40 - M)$$

Because $M(t) < 40$, $\frac{d^2M}{dt^2} < 0$, so the graph of M is concave down. Therefore, the tangent line approximation of $M(2)$ is an overestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 4 points.

A response that presents an antiderivative of $-\ln(40 - M)$ without absolute value symbols is eligible for all 4 points.

A response with no constant of integration can earn at most the first 2 points.

A response is eligible for the third point only if it has earned the first 2 points.

7-7

- Special Case: A response that presents $+\ln(40-M)=\frac{t}{4}+C$ (or equivalent) does not earn the second point, is eligible for the third point, but not eligible for the fourth.

An eligible response earns the third point by correctly including the constant of integration in an equation and substituting 0 for t and 5 for M.

A response is eligible for the fourth point only if it has earned the first 3 points.

A response earns the fourth point only for an answer of $M=40-35e^{-\frac{t}{4}}$ or equivalent.

0	1	2	3	4 ✓

The student response accurately includes **all four** of the criteria below.

- ☐ Separates variables
- ☐ Finds antiderivatives
- ☐ Constant of integration and uses initial condition
- ☐ Solves for M

Solution:

$$\frac{dM}{40-M}=\frac{1}{4}dt$$

$$\int\frac{dM}{40-M}=\int\frac{1}{4}dt$$

$$-\ln|40-M|=\frac{1}{4}t+C$$

$$-\ln|40-5|=0+C\Rightarrow C=-\ln 35$$

$$M(t)<40\Rightarrow 40-M>0\Rightarrow|40-M|=40-M$$

$$-\ln(40-M)=\frac{1}{4}t-\ln 35$$

$$\ln(40-M)=-\frac{1}{4}t+\ln 35$$

$$40-M=35e^{-\frac{t}{4}}$$

$$M=40-35e^{-\frac{t}{4}}$$

7-7

38. Which of the following is the particular solution to the differential equation $\frac{dy}{dx} = \sin\left(x^2\right)$ with the initial condition $y\left(\sqrt{\pi}\right) = 4$?

(A) $y = -\cos\left(x^2\right) + 3$

(B) $y = -\frac{\cos\left(x^2\right)}{2x} + 4 - \frac{1}{2\sqrt{\pi}}$

(C) $y = 4 + \int_0^x \sin\left(t^2\right)\,dt$

(D) $y = 4 + \int_{\sqrt{\pi}}^x \sin\left(t^2\right)\,dt$

Answer D

Correct. The function F defined by $F(x) = y_0 + \int_a^x f(t)\,dt$ is a particular solution to the differential equation $\frac{dy}{dx} = f(x)$ satisfying $F(a) = y_0$.

39. Which of the following is the solution to the differential equation $\frac{dy}{dx} = y^2 - (xy)^2$ with the initial condition $y(3) = 1$?

(A) $y = \frac{5}{6} - \frac{3}{3x - x^3}$

(B) $y = \frac{-3}{3x - x^3 + 15}$

(C) $y = \left(3x - x^3 + 19\right)^{\frac{1}{3}}$

(D) $y = \left(e^{x - \frac{1}{3}x^3 + 6}\right)^{\frac{1}{2}}$

Answer B

Correct. This question involves solving a differential equation by separation of variables and using the initial condition to determine the appropriate value for the arbitrary constant.

$\frac{dy}{dx} = y^2 - (xy)^2 = y^2 - x^2y^2 = y^2\left(1 - x^2\right) \Rightarrow \frac{1}{y^2}\,dy = \left(1 - x^2\right)dx$

$\int \frac{1}{y^2}\,dy = \int \left(1 - x^2\right)dx \Rightarrow -\frac{1}{y} = x - \frac{1}{3}x^3 + C$

$-1 = 3 - \frac{1}{3}(3)^3 + C = 3 - 9 + C \Rightarrow C = 5$

$-\frac{1}{y} = x - \frac{1}{3}x^3 + 5 = \frac{3x - x^3 + 15}{3} \Rightarrow y = \frac{-3}{3x - x^3 + 15}$

7-7

40. The general solution to the differential equation $\frac{dy}{dx} = \frac{x}{y}$ is $y = \pm\sqrt{x^2 + C}$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(-5) = -4$. Which of the following is an expression for $f(x)$ and its domain?

(A) $f(x) = -\sqrt{x^2 - 9}$ for $x < -3$ ✓

(B) $f(x) = \sqrt{x^2 - 9}$ for $x < -3$

(C) $f(x) = -\sqrt{x^2 - 9}$ for $|x| > 3$

(D) $f(x) = \sqrt{x^2 - 9}$ for $|x| > 3$

Answer A

Correct. The choice of the positive or negative sign for the square root when solving for y and the choice of the interval for the domain of the particular solution are made based on the initial value.
The initial y-value determines which sign to use for the square root. Since the initial y-value is negative, the particular solution uses the negative sign for the square root.
The domain of a particular solution to a differential equation is the largest open interval containing the initial value on which the solution is differentiable and satisfying the differential equation. Since $x^2 - 9 > 0$ for $|x| > 3$ and the initial x-value is -5, the domain for the particular solution is the interval $x < -3$. The graph of the continuous solution cannot cross over the gap between $x = -3$ and $x = 3$ where the expression $-\sqrt{x^2 - 9}$ is not defined.

41. Which of the following is the particular solution to the differential equation $\frac{dy}{dx} = \sqrt{x^9 + 1}$ with the initial condition $y(1) = 3$?

(A) $y = \frac{2}{3}\left(x^9 + 1\right)^{\frac{3}{2}} + \frac{7}{3}$

(B) $y = \left(\frac{2}{27x^8}\right)\left(x^9 + 1\right)^{\frac{3}{2}} + \frac{81 - 4\sqrt{2}}{27}$

(C) $y = 3 + \int_0^x \sqrt{t^9 + 1}\ dt$

(D) $y = 3 + \int_1^x \sqrt{t^9 + 1}\ dt$ ✓

Answer D

Correct. The function F defined by $F(x) = y_0 + \int_a^x f(t)\,dt$ is a particular solution to the differential equation $\frac{dy}{dx} = f(x)$ satisfying $F(a) = y_0$.

7-7

42. Which of the following is the solution to the differential equation $\frac{dy}{dx} = (x-2)(y-2)$ for $y > 2$ with the initial condition $y(4) = 5$?

(A) $y = 4 + e^{\left(\frac{x^2}{2} - 2x\right)}$

(B) $y = 2 + 3e^{\left(\frac{x^2}{2} - 2x\right)}$

(C) $y = \frac{4+\sqrt{16+4(x^2-4x+5)}}{2}$

(D) $y = \frac{-x^2+4x+5}{\left(1-\frac{x^2}{2}+2x\right)}$

Answer B

Correct. This question involves solving a differential equation by separation of variables and using the initial condition to determine the appropriate value for the arbitrary constant.

$\frac{dy}{dx} = (x-2)(y-2)$

Separating the variables gives

$\frac{dy}{y-2} = (x-2)\,dx.$

Then $\displaystyle\int \frac{1}{y-2}\,dy = \int (x-2)\,dx.$

$\ln|y-2| = \frac{x^2}{2} - 2x + C$

Since $y > 2$, $(y-2) = e^{\left(\frac{x^2}{2} - 2x + C\right)} = Ae^{\left(\frac{x^2}{2} - 2x\right)}.$

$y = 2 + Ae^{\left(\frac{x^2}{2} - 2x\right)}$

Since $y(4) = 5$,

$5 = 2 + Ae^0 \Rightarrow A = 3.$

Then $y = 2 + 3e^{\left(\frac{x^2}{2} - 2x\right)}.$

7-7

43. The general solution to the differential equation $\frac{dy}{dx} = \frac{x}{y}$ is $y = \pm\sqrt{x^2 + C}$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(-2) = -\sqrt{2}$. Which of the following is an expression for $f(x)$ and the domain for which the solution is valid?

(A) $f(x) = -\sqrt{x^2 - 2}$ for $x < -\sqrt{2}$ ✓

(B) $f(x) = \sqrt{x^2 - 2}$ for $x < -\sqrt{2}$

(C) $f(x) = -\sqrt{x^2 - 2}$ for $|x| > \sqrt{2}$

(D) $f(x) = \sqrt{x^2 - 2}$ for $|x| > \sqrt{2}$

Answer A

Correct. The choice of the positive or negative sign for the square root when solving for y, and the choice of the interval for the domain of the particular solution are made based on the initial value. The initial y-value determines which sign to use for the square root. Since the initial y-value is negative, the particular solution uses the negative sign for the square root. The domain of a particular solution to a differential equation is the largest open interval containing the initial value on which the solution is differentiable and satisfies the differential equation. Since $x^2 - 2 > 0$ for $|x| > \sqrt{2}$ and the initial x-value is -2, the domain for the particular solution is the interval $x < -\sqrt{2}$. The graph of the continuous solution cannot cross over the gap between $x = -\sqrt{2}$ and $x = \sqrt{2}$ where the expression $-\sqrt{x^2 - 2}$ is not defined.

44. If $\frac{dy}{dx} = 16\sin^3 x \cos x$ and $y = 16$ when $x = \frac{\pi}{2}$, what is the value of y when $x = \frac{\pi}{6}$?

(A) $\frac{49}{4}$ ✓

(B) $\frac{1}{4}$

(C) 10

(D) 13

Answer A

Correct. The differential equation can be solved using antidifferentiation and the initial condition to determine the constant of integration. The solution can then be evaluated at $x = \frac{\pi}{6}$.

Starting with the substitution $u = \sin x$, $\frac{du}{dx} = \cos x \Rightarrow du = \cos x\, dx$.

$$y = \int 16\sin^3 x \cos x\, dx = \int 16u^3\, du = 4u^4 + C = 4\sin^4 x + C$$

$$16 = 4\sin^4\left(\frac{\pi}{2}\right) + C = 4 + C \Rightarrow C = 12$$

When $x = \frac{\pi}{6}$, $y = 4\sin^4\left(\frac{\pi}{6}\right) + 12 = 4\left(\frac{1}{2}\right)^4 + 12 = \frac{1}{4} + 12 = \frac{49}{4}$.

7-7

45. Which of the following is the solution to the differential equation $\frac{dP}{dt} + P = 10$ with the initial condition $P(0) = 4$?

(A) $P = -1 + \sqrt{20t + 25}$

(B) $P = 5 - e^{-t}$

(C) $P = 10 - 6e^{-t}$ ✓

(D) $P = 10 - 6e^{t}$

Answer C

Correct. This is a separable differential equation since it only contains the dependent variable. Rewriting the differential equation as $\frac{dP}{dt} = 10 - P$ allows for the separation, as follows.

$\frac{dP}{dt} = 10 - P \Rightarrow \frac{dP}{10-P} = dt$

$$\int \frac{1}{10 - P}\, dP = \int 1\, dt \Rightarrow -\ln|10 - P| = t + C$$

$-\ln|10 - P| = t + C \Rightarrow \ln|10 - P| = -t - C \Rightarrow |10 - P| = e^{-t-C} = e^{-C}e^{-t} = Ae^{-t}$

$10 - P = \pm Ae^{-t} = Be^{-t}$

$10 - 4 = Be^{0} = B \Rightarrow B = 6$

$10 - P = 6e^{-t} \Rightarrow P = 10 - 6e^{-t}$

46. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = \frac{e^x - 1}{e^y}$ with the initial condition $f(1) = 0$. What is the value of $f(-2)$?

(A) 0.217

(B) 0.349 ✓

(C) 0.540

(D) 0.759

Answer B

Correct. This question involves solving a differential equation by separation of variables and using the initial condition to determine the appropriate value of the arbitrary constant.

$\frac{dy}{dx} = \frac{e^x - 1}{e^y} \Rightarrow e^y dy = (e^x - 1)dx$

$$\int e^y dy = \int (e^x - 1)dx \Rightarrow e^y = e^x - x + C$$

$e^0 = e^1 - 1 + C \Rightarrow 1 = e - 1 + C \Rightarrow C = 2 - e$

$e^y = e^x - x + 2 - e \Rightarrow y = \ln(e^x - x + 2 - e)$

$f(-2) = \ln\left(e^{-2} + 2 + 2 - e\right) \approx 0.349$

7-7

47. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = \frac{x^2+1}{e^y}$ with the initial condition $f(1) = 0$. What is the value of $f(2)$?

(A) 1.253

(B) 1.466 ✓

(C) 2.197

(D) 2.303

Answer B

Correct. This question involves solving a differential equation by separation of variables and using the initial condition to determine the appropriate value of the arbitrary constant.

$\frac{dy}{dx} = \frac{x^2+1}{e^y} \Rightarrow e^y dy = (x^2+1)dx$

$\int e^y dy = \int (x^2+1)dx \Rightarrow e^y = \frac{x^3}{3} + x + C$

$f(1) = 0 \Rightarrow e^0 = \frac{1}{3} + 1 + C \Rightarrow C = 1 - \frac{4}{3} = -\frac{1}{3}$

$e^y = \frac{x^3}{3} + x - \frac{1}{3} \Rightarrow y = \ln\left(\frac{x^3}{3} + x - \frac{1}{3}\right)$

$f(2) = \ln\left(\frac{8}{3} + 2 - \frac{1}{3}\right) = 1.466$

48. Which of the following is the solution to the differential equation $\frac{dy}{dt} - 2 = -y$ with the initial condition $y(0) = -3$?

(A) $y = -1 - 2\sqrt{t+1}$

(B) $y = -e^{-t} - 2$

(C) $y = 2 - 5e^{-t}$ ✓

(D) $y = 2 - 5e^{t}$

Answer C

Correct. This is a separable differential equation since it only contains the dependent variable. Rewriting the differential equation as $\frac{dy}{dt} = 2 - y$ allows for the separation, as follows.

$\frac{dy}{dt} = 2 - y \Rightarrow \frac{1}{2-y}dy = dt$

$\int \frac{1}{2-y}dy = \int 1\,dt \Rightarrow \ln|2-y| = -t + C \Rightarrow |2-y| = e^{-t+C} = Ae^{-t}$

$y(0) = -3 \Rightarrow |2-y| > 0 \Rightarrow 2 - y = Ae^{-t} \Rightarrow y = 2 - Ae^{-t}$

$y(0) = 3 \Rightarrow -3 = 2 - Ae^0 = 2 - A \Rightarrow A = 5 \Rightarrow y = 2 - 5e^{-t}$

7-7

49. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = y^2$ with the initial condition $f(1) = 1$. Which of the following gives an expression for $f(x)$ and its domain?

(A) $f(x) = \frac{2x-1}{x}$ for $x \neq 0$

(B) $f(x) = \frac{2x-1}{x}$ for $x > 0$

(C) $f(x) = \frac{1}{2-x}$ for $x \neq 2$

(D) $f(x) = \frac{1}{2-x}$ for $x < 2$

Answer D

Correct. This question involves solving a differential equation by separation of variables and using the initial condition to determine the appropriate value of the arbitrary constant. The solution to the differential equation may be subject to domain restrictions.

$\frac{dy}{dx} = y^2 \Rightarrow \frac{1}{y^2}dy = 1dx$

$$\int \frac{1}{y^2}dy = \int \frac{1}{y^2}dy \Rightarrow -\frac{1}{y} = x + C$$

$-\frac{1}{1} = 1 + C \Rightarrow C = -2$

$-\frac{1}{y} = x - 2 \Rightarrow y = \frac{1}{2-x}$

The domain of a particular solution to a differential equation is the largest open interval containing the initial value on which the solution is differentiable and satisfies the differential equation. This solution is not defined at $x = 2$ and hence not differentiable there. Since the initial condition is at $x = 1$, the domain of this solution is $x < 2$. The graph of the continuous solution cannot cross the vertical asymptote at $x = 2$.

7-7

50. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = \frac{1}{2y+1}$ with the initial condition $y(0) = 0$. Which of the following gives an expression for $f(x)$ and the domain for which the solution is valid?

(A) $y = \frac{e^{2x}-1}{2}$ for all x

(B) $y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

(C) $y = \sqrt{x}$ for $x \geq 0$

(D) $y = \frac{-1+\sqrt{1+4x}}{2}$ for $x > -\frac{1}{4}$

Answer D

Correct. This question involves solving a differential equation by separation of variables and using the initial condition to determine the appropriate value of the arbitrary constant. The solution to the differential equation may be subject to domain restrictions.

$\frac{dy}{dx} = \frac{1}{2y+1}$

$\int (2y+1)\, dy = \int 1\, dx$

$y^2 + y = x + C$

$y^2 + y - (x + C) = 0$

$y = \frac{-1 \pm \sqrt{1+4(x+C)}}{2}$

Since $y(0) = 0$, $0 = \frac{-1 \pm \sqrt{1+4C}}{2} \Rightarrow -1 \pm \sqrt{1+4C} = 0 \Rightarrow$
$\pm\sqrt{1+4C} = 1$.

Thus, the positive square root is taken and

$\sqrt{1+4C} = 1 \Rightarrow C = 0$.

Thus, $y = \frac{-1+\sqrt{1+4x}}{2}$. The domain of a particular solution to a differential equation is the largest open interval containing the initial value on which the solution is differentiable and satisfies the differential equation. Since $1 + 4x > 0$ for $x > -\frac{1}{4}$ and the initial x-value is 0, the domain for the particular solution is the interval $x > -\frac{1}{4}$.

7-7

51. If $\frac{dx}{dt} = 4t^3$ and if $x = 5$ when $t = 1$, what is the value of x when $t = 2$?

(A) 12

(B) 16

(C) 20

(D) 41

Answer C

Correct. The power rule for antiderivatives is used to find the general solution for x, and the initial condition is used to find the constant of integration, as follows.

$x = t^4 + C$

When $t = 1$, $x = 5$ so $5 = 1 + C$ and $C = 4$.

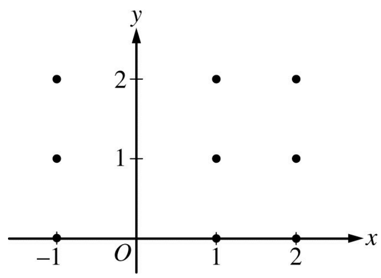
When $t = 2$, $x = 2^4 + 4 = 16 + 4 = 20$.

7-7

52. Consider the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y-1)$, where $x \neq 0$. Let $y = f(x)$ be the particular solution to the differential equation with initial condition $f(1) = 2$.

(a) Find the slope of the line tangent to the graph of f at the point $(1, 2)$.

(b) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



(c) Find the particular solution $y = f(x)$ to the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y-1)$ with initial condition $f(1) = 2$.

Part A

1 point is earned for the correct answer

$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2-1) = -1$$

0	1

The student earns all of the following point:

1 point is earned for the correct answer

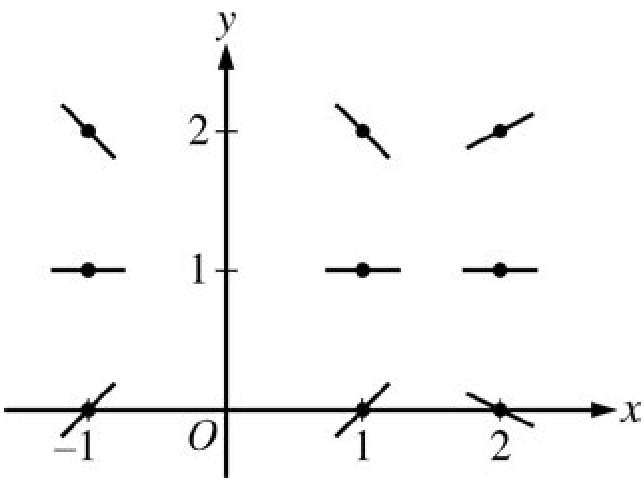
$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2-1) = -1$$

Part B

1 point is earned for zero slope at each point (x, y) where $y = 1$

1 point is earned for the remaining slopes

7-7

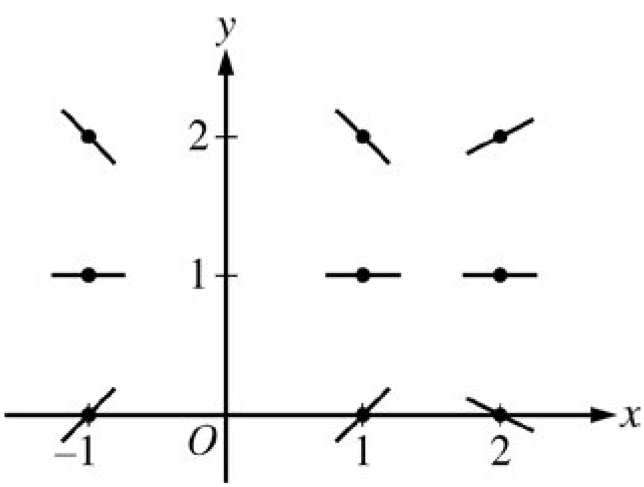


0	1	2

The student earns all of the following points:

1 point is earned for zero slope at each point (x, y) where $y = 1$

1 point is earned for the remaining slopes



Part C

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses of the initial condition

1 point is earned for the solution for y

7-7

$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y-1)$$
$$\int \frac{dy}{y-1} = \int \left(1 - \frac{2}{x^2}\right) dx$$
$$\ln|y-1| = x + \frac{2}{x} + C$$
$$\ln|2-1| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$
$$\ln|y-1| = x + \frac{2}{x} - 3$$

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

$$\ln(y-1) = x + \frac{2}{x} - 3$$
$$y = f(x) = e^{\left(x + \frac{2}{x} - 3\right)} + 1$$

Note: This solution is valid for $x > 0$.

Note: max 3/6 [1-2-0-0-0] if no constant of integration.

Note: 0/6 if no separation of variables

0	1	2	3	4	5	6

The student earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses of the initial condition

1 point is earned for the solution for y

$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y-1)$$
$$\int \frac{dy}{y-1} = \int \left(1 - \frac{2}{x^2}\right) dx$$
$$\ln|y-1| = x + \frac{2}{x} + C$$
$$\ln|2-1| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$
$$\ln|y-1| = x + \frac{2}{x} - 3$$

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

7-7

$$\ln(y-1) = x + \frac{2}{x} - 3$$
$$y = f(x) = e^{\left(x+\frac{2}{x}-3\right)} + 1$$

Note: This solution is valid for $x > 0$.

Note: max 3/6 [1-2-0-0-0] if no constant of integration.

Note: 0/6 if no separation of variables